ABBREVIATIONS

A/C	ABOVE CEILING
AC	AIR CONDITIONING
AD	ACCESS DOOR
ADJ	ADJUSTABLE
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AUTO	AUTOMATIC
B/G	BELOW GRADE
B/S	BELOW SLAB
BAL	BALANCING
BCO	BASE CLEANOUT
BDD	BACKDRAFT DAMPER
BFLY	BUTTERFLY
BHP	BRAKE HORSEPOWER
BOD	BASIS OF DESIGN
BOD	BOTTOM OF DUCT
CD	CEILING DIFFUSER
CFM	CUBIC FEET PER MINUTE
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER SUPPLY
CO	CLEANOUT
COND	CONDENSATE
CSR	CURVED SUPPLY REGISTER
CU	CONDENSING UNIT
CW	COLD WATER (DOMESTIC)
CWR	CONDENSER WATER RETURN
CWS	CONDENSER WATER SUPPLY
dB	DECIBELS
DB	DRY BULB
DN	DOWN
DR	DRAIN
DWG	DRAWING
EA	EACH
EAT	ENTERING AIR TEMP.
ECC	ECCENTRIC
EF	EXHAUST FAN
EFF	EFFICIENCY
EG	EXHAUST GRILLE
ER	EXHAUST REGISTER
ESP	EXTERNAL STATIC PRESSURE
EWT	ENTERING WATER TEMP.
EX	EXISTING
EXH	EXHAUST
F	FAHRENHEIT
FCO	FLOOR CLEANOUT
FCU	FAN COIL UNIT
FD	FIRE DAMPER
FLR	FLOOR
FOB	FLAT ON BOTTOM
FOR	FUEL OIL RETURN
FOS	FUEL OIL SUPPLY
FOT	FLAT ON TOP
FPB	FAN POWERED BOX
FPM	FEET PER MINUTE
FPS	FEET PER SECOND
FSD	FIRE/SMOKE DAMPER
FT	FEET
G	GATE
GA	GAUGE
GPM	GALLONS PER MINUTE
HD	HUB DRAIN
HP	HORSEPOWER
HTG	HEATING
HW	HOT WATER (DOMESTIC)
HWR	HEATING HOT WATER RETURN
HWS	HEATING HOT WATER SUPPLY
HWRR	HOT WATER REVERSE RETURN
ID	INSIDE DIMENSION
IN	INCHES
KW	KILOWATTS
LAT	LEAVING AIR TEMPERATURE
LB	POUNDS
LSD	LINEAR SLOT DIFFUSER
LWR	LOOP WATER RETURN
LWS	LOOP WATER SUPPLY
MAX	MAXIMUM
MD	MANUAL DAMPER
MIN	MINIMUM
MOD	MOTOR OPERATED DAMPER
MFR	MANUFACTURER
NC	NORMALLY CLOSED
NG	NATURAL GAS
NO	NORMALLY OPEN
NOM	NOMINAL
OA	OUTSIDE AIR
OBD	OPPOSED BLADE DAMPER
OD	OUTSIDE DIMENSION
PIU	POWERED INDUCTION UNIT
PSI	POUNDS PER SQUARE INCH
RA	RETURN AIR
RAD	RADIUS
RAG	RETURN AIR GRILLE
RAR	RETURN AIR REGISTER
RED	REDUCER
RL	REFRIGERANT LIQUID
RS	REFRIGERANT SUCTION
RTU	ROOFTOP UNIT
SA	SUPPLY AIR
SAN	SANITARY
SD	SMOKE DAMPER
SD	SLOT DIFFUSER
SEN	SENSIBLE
SP	STATIC PRESSURE
SQ	SQUARE
SR	SUPPLY REGISTER
SS	SPLIT SYSTEM
ST	STORM
TEMP	TEMPERATURE
TG	TRANSFER GRILLE
TYP	TYPICAL
UON	UNLESS OTHERWISE NOTED
V	VENT
VA	VALVE
VAV	VARIABLE AIR VOLUME
VTR	VENT THRU ROOF
wb	WET BULB
WC	WATER COLUMN
WHA	WATER HAMMER ARRESTOR
WT	WEIGHT
W	WASTE

APPLICABLE CODES

- International Building Code, 2024 Edition, with Georgia Amendments (2026)
- International Fire Code, 2024 Edition
- International Mechanical Code, 2024 Edition, with Georgia Amendments (2026)
- International Fuel Gas Code, 2024 Edition, with Georgia Amendments (2026)
- International Energy Conservation Code, 2015 Edition, with Georgia Supplements and Amendments

MECHANICAL SHEET INDEX:

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M-501	MECHANICAL DETAILS
M-502	MECHANICAL DETAILS AND CONTROLS
M-503	MECHANICAL CONTROLS DETAILS

SPECIFICATIONS

PART 1 - GENERAL

1. SUBMITTALS
   A. PRODUCT DATA: FOR EACH TYPE OF PRODUCT INDICATED.

2. WARRANTY WARRANT NEW AND MODIFIED DUCTWORK FOR PERIOD OF ONE YEAR FROM DATE OF FINAL ACCEPTANCE OF JOB, AGAINST NOISE AND VIBRATION UNDER FULL RANGE OF OPERATING CONDITIONS.

PART 2 - PRODUCTS

1. METAL DUCTS
   A. CONTINUOUS HOT-DIP MILL GALVANIZED. MINIMUM COATING OF G60, LOCK-FORMING QUALITY STEEL SHEETS IN ACCORDANCE WITH ASTM A653.
   B. GAUGES: SHEET METAL GAUGE AS SPECIFIED IN SMACNA - HVAC DUCT CONSTRUCTION STANDARDS BUT NOT LESS THAN THE FOLLOWING:
      - RECTANGULAR DUCT: 26 GAUGE FOR ALL SIZES.
      - ROUND DUCT: PRIME GRADE STEEL SHEET.
        a. 14 INCH DIAMETER AND SMALLER: 26 GAUGE.
        b. 15 INCH DIAMETER THRU 26 INCH DIAMETER: 24 GAUGE.
        c. ROUND DUCT SHALL BE SPIRAL SEAM WHERE EXPOSED AND NOT INSULATED. IN ALL OTHER AREAS SPIRAL OR LONGITUDINAL SHALL BE USED.

2. TURNING VANES
   A. MANUFACTURED TURNING VANES FOR METAL DUCTS: CURVED BLADES OF GALVANIZED SHEET STEEL; SUPPORT WITH BARS PERPENDICULAR TO BLADES SET; SET INTO VANE RUNNERS SUITABLE FOR DUCT MOUNTING.
   B. GENERAL REQUIREMENTS: COMPLY WITH SMACNA "HVAC DUCT CONSTRUCTION STANDARDS - METAL AND FLEXIBLE"; FIGURES 4-3 "VANES AND VANE RUNNERS," 4-4, "VANE SUPPORT IN ELBOWS."
   C. VANE CONSTRUCTION: SINGLE WALL

3. SEALANT AND GASKET

   A. GENERAL SEALANT AND GASKET REQUIREMENTS: SURFACE-BURNING CHARACTERISTICS FOR SEALANTS AND GASKETS SHALL BE A MAXIMUM FLAME-SPREAD INDEX OF 25 AND A MAXIMUM SMOKE- DEVELOPED INDEX OF 50 WHEN TESTED ACCORDING TO UL 723; CERTIFIED BY AN NRTL.

4. HANGERS AND SUPPORTS

   A. HANGER RODS: CADMIUM-PLATED STEEL RODS AND NUTS.
   B. STRAP AND ROD SIZES: COMPLY WITH SMACNA'S "HVAC DUCT CONSTRUCTION STANDARDS - METAL AND FLEXIBLE" TABLE 5-1." RECTANGULAR DUCT HANGERS MINIM SIZE." AND TABLE 5-2 "MINIMUM HANGER SIZES FOR ROUND DUCT."
   C. STEEL CABLES FOR GALVANIZED STEEL DUCTS: GALVANIZED STEEL COMPLYING WITH ASTM A 603.
   D. STEEL CABLE END CONNECTIONS: CADMIUM PLATED STEEL ASSEMBLIES WITH BRACKETS, SWIVEL, AND BOLTS DESIGNED FOR DUCT HANGER SERVICE; WITH AN AUTOMATIC-LOCKING AND CLAMPING DEVICE.
   E. DUCT ATTACHMENTS: SHEET METAL SCREWS, BLIND RIVETS, OR SELF-TAPPING METAL SCREWS; COMPATIBLE WITH DUCT MATERIALS.

5. MECHANICAL EQUIPMENT

   A. PROVIDE EQUIPMENT INDICATED ON DRAWINGS.
   B. PRODUCTS REQUIRING ELECTRICAL CONNECTION: LISTED AND CLASSIFIED BY UNDERWRITERS LABORATORIES, INCORPORATED, AND ACCEPTABLE TO THE AUTHORITY HAVING JURISDICTION AS SUITABLE FOR THE PURPOSE SPECIFIED AND INDICATED.
   C. OPERATION AND MAINTENANCE DATA: INCLUDE MANUFACTURER'S DESCRIPTIVE LITERATURE. OPERATING INSTRUCTIONS, MAINTENANCE AND REPAIR DATA, AND PARTS LISTING FOR MECHANICAL EQUIPMENT.

6. CONTROLS

   A. PROVIDE COMPLETE CONTROLS SYSTEM INCLUDING STEP-DOWN TRANSFORMERS FOR 24 VAC CONTROL, RELAYS, RELAY BOARDS, RELAY BASES, PRINTED CIRCUIT BOARDS, THERMOSTATS, TIMERS, SENSORS, INDICATORS, ACTUATORS, AND OTHER NECESSARY DEVICES REQUIRED FOR COMPLETE INSTALLATION INCLUDING ENCLOSURES FOR EQUIPMENT. LOCATIONS AND TYPES OF THERMOSTATS, TIMERS, AND OTHER CONTROLS EQUIPMENT ARE INDICATED ON DRAWINGS.
   B. POWER SUPPLIES: TRANSFORMERS WITH CLASS 2 CURRENT-LIMITING TYPE OR OVERCURRENT PROTECTION; LIMIT CONNECTED LOADS TO 80 PERCENT OF RATED CAPACITY. DC POWER SUPPLY SHALL MATCH OUTPUT CURRENT AND VOLTAGE REQUIREMENTS AND BE FULL WAVE RECTIFIER TYPE.
   C. DEVICES
      - TEMPERATURE SENSORS
        a. ACCURACY: PLUS OR MINUS 0.36 DEG F OR 0.2 PERCENT AT CALIBRATION POINT.
        b. WIRE: TWISTED SHIELDED PAIR - CABLE.
        c. OUTSIDE AIR SENSORS: WATERTIGHT INLET FITTING, SHIELDED FROM DIRECT SUNLIGHT.
      - PROGRAMMABLE THERMOSTAT
        a. ELECTRIC, SOLID STATE, MICROCOMPUTER-BASED ROOM THERMOSTAT WITH REMOTE SENSOR.
        b. AUTOMATIC SWITCHING FROM HEATING TO COOLING.
      - LOW VOLTAGE ON / OFF THERMOSTATS
        a. MC TO INSTALL A TEMPORARY DIGITAL THERMOSTAT FOR A START-UP. FINAL THERMOSTAT TO BE KMC FLEXSTAT PROVIDED AND INSTALLED BY CONTROLS VENDOR.
   D. LOW VOLTAGE CONTROL WIRING
      - JACKETED, INDIVIDUALLY SHIELDED TWISTED PAIRS OF STRANDED INSULATED TINNED COPPER CONDUCTORS.
      - STRANDED TINNED COPPER DRAIN WIRE.
      - PLENUM-RATED JACKET FOR PLENUM APPLICATIONS.
      - CONDUCTOR SIZES BASED ON 10% MAXIMUM VOLTAGE DROP.
      - MINIMUM WIRE SIZE: 22 AWG, MAXIMUM WIRE SIZE: 14 GAUGE.

PART 3 - EXECUTION

1. GENERAL

   A. UNLESS OTHERWISE DICTATED BY APPLICABLE CODES OR THE AUTHORITY HAVING JURISDICTION, ALL MATERIALS WITHIN RETURN AIR PLENUM MUST BE NONCOMBUSTIBLE AND/OR HAVE A FLAME SPREAD INDEX OF NOT MORE THAN 25 AND A SMOKE-DEVELOPED INDEX NOT MORE THAN 50 IN ACCORDANCE WITH ASTM E 84 TESTING.
   B. PROTECT WORK, EQUIPMENT AND MATERIAL TO PREVENT OBSTRUCTION, DAMAGE OR BREAKAGE. CLOSE PIPE OPENINGS WITH CAPS OR PLUGS DURING INSTALLATION. COVER AND PROTECT EQUIPMENT AGAINST DIRT, WATER, CHEMICAL OR MECHANICAL INJURY. AT THE COMPLETION OF WORK, THOROUGHLY CLEAN ALL EQUIPMENT AND DELIVER THE ENTIRE SYSTEM IN AN UNBLEMISHED CONDITION.
   C. MAKE CHANGES IN PULLEYS, BELTS, DUCTWORK, AND DAMPERS AS REQUIRED FOR CORRECT BALANCE AS RECOMMENDED BY AIR BALANCE AGENCY.
   D. PROVIDE PENETRATION FIRESTOPPING THAT IS PRODUCED AND INSTALLED TO MAINTAIN ORIGINAL FIRE-RESISTANCE RATING OF CONSTRUCTION PENETRATED. PENETRATION FIRESTOPPING SYSTEMS SHALL BE COMPATIBLE WITH ONE ANOTHER, WITH THE SUBSTRATES FORMING OPENING, AND WITH PENETRATING ITEMS IF ANY.
      - COORDINATE SIZING OF SLEEVES, OPENINGS CORE-DRILLED HOLES, OR CUT OPENINGS TO ACCOMMODATE PENETRATION FIRESTOPPING.
      - INSTALL PENETRATION FIRESTOPPING TO COMPLY WITH MANUFACTURER'S WRITTEN INSTALLATION INSTRUCTIONS AND PUBLISHED DRAWINGS FOR PRODUCTS AND APPLICATIONS INDICATED.

2. DUCT INSTALLATION
   A. INSTALL DUCTS ACCORDING TO SMACNA STANDARDS.
   B. PROTECT DUCT INTERIORS FROM MOISTURE, CONSTRUCTION DEBRIS, AND OTHER FOREIGN MATERIALS.
   C. FABRICATE, ERECT, AND INSTALL DUCTWORK FOR HEATING, VENTILATING, AND AIR CONDITIONING PER SMACNA STANDARDS AND THE REQUIREMENTS OF THE AUTHORITY HAVING JURISDICTION.
   D. MAKE JOINTS AIR TIGHT ON ALL DUCTWORK. SEAL EXHAUST DUCTWORK WITH EXTERIOR SEAL AS REQUIRED TO ASSURE POSITIVE SEAL. COMPLY WITH SMACNA DUCT CLASS A.
   E. INSTALL FACTORY FABRICATED FITTINGS FOR CHANGES IN DIRECTION, SIZE, AND SHAPE AND FOR BRANCH CONNECTIONS.

3. HANGERS AND SUPPORT INSTALLATION
   A. COMPLY WITH SMACNA STANDARDS.
   B. INSTALL HANGERS AND SUPPORTS WITHIN 24" OF EACH ELBOW AND WITH 48" OF EACH BRANCH INTERSECTION.
   C. DO NOT USE FASTENERS THAT PENETRATE ROOF DECK.

4. MECHANICAL EQUIPMENT
   A. INSTALL PER MANUFACTURER'S RECOMMENDATIONS.

5. CONTROLS
   A. COORDINATE LOCATION OF THERMOSTATS, HUMIDISTATS, AND OTHER EXPOSED CONTROL SENSORS WITH PLANS AND ROOM DETAILS BEFORE INSTALLATION.
   B. CONNECT AND CONFIGURE EQUIPMENT AND SOFTWARE TO ACHIEVE SEQUENCE OF OPERATIONS.
   C. LOW VOLTAGE CONTROL WIRING.
      - COMPLY WITH NECA 1.
      - COMPLY WITH TIA/EIA-568-B.1.
      - BUNDLE AND HARNESS MULTICONDUCTOR INSTRUMENT CABLE IN PLACE OF SINGLE CABLES WHERE SEVERAL CABLES FOLLOW A COMMON PATH.
      - NUMBER-CODE OR COLOR-CODE CONDUCTORS FOR FUTURE IDENTIFICATION AND SERVICE OF CONTROL SYSTEM, EXCEPT LOCAL INDIVIDUAL ROOM CONTROL CABLES.
      - CONCEAL CABLE, EXCEPT IN MECHANICAL ROOMS AND AREAS WHERE OTHER CONDUIT AND PIPING ARE EXPOSED.
      - INSTALL PLENUM-RATED CABLE IN ENVIRONMENTAL AIR SPACES INCLUDING PLENUM CEILINGS.
      - WIRING WITHIN ENCLOSURES
        a. BUNDLE, LACE, AND TRAIN CONDUCTORS TO TERMINAL POINTS WITH NO EXCESS.
        b. INSTALL CONDUCTORS PARALLEL, WITH OR AT RIGHT ANGLES TO SIDES AND BACK OF ENCLOSURE.
   D. AFTER ELECTRICAL CIRCUITRY HAS BEEN ENERGIZED, START UNITS TO CONFIRM PROPER UNIT OPERATION. REMOVE AND REPLACE MALFUNCTIONING UNITS AND RETEST.
   E. TEST EACH SYSTEM FOR COMPLIANCE WITH SEQUENCE OF OPERATION.
   F. OCCUPANCY ADJUSTMENTS, WHEN REQUESTED WITHIN 12 MONTHS OF DATE OF SUBSTANTIAL COMPLETION, PROVIDE ON-SITE ASSISTANCE IN ADJUSTING SYSTEM TO SUIT ACTUAL OCCUPIED CONDITIONS. PROVIDE UP TO THREE VISITS TO PROJECT DURING OTHER THAN NORMAL OCCUPANCY HOURS FOR THIS PURPOSE.

6. EQUIPMENT LABEL INSTALLATION

   A. INSTALL OR PERMANENTLY FASTEN LABELS ON EACH MAJOR ITEM OF MECHANICAL EQUIPMENT.
   B. LABEL ACCESS DOORS TO INDICATE THE PURPOSE OF ACCESS DOOR.
   C. LOCATE EQUIPMENT LABELS WHERE ACCESSIBLE AND VISIBLE.
   D. WHERE EQUIPMENT, SUCH AS A VAV BOX, IS CONCEALED BY ACCESSIBLE DROP OR LAY-IN CEILINGS OR OTHER ACCESSIBLE BARRIER, ATTACH LABEL TO CEILING GRID OR A VISIBLE LOCATION TO IDENTIFY LOCATION OF HIDDEN EQUIPMENT.

7. TESTING, ADJUSTING, AND BALANCING

   A. PERFORM TESTING AND BALANCING PROCEDURES ON EACH SYSTEM ACCORDING TO THIS SECTION AND THE PROCEDURES CONTAINED IN ONE OF THE FOLLOWING:
      - AABC'S "NATIONAL STANDARDS FOR TOTAL SYSTEM BALANCE"
      - NEBB'S "PROCEDURAL STANDARDS FOR TESTING, ADJUSTING, AND BALANCING OF ENVIRONMENTAL SYSTEMS."
   B. COMPLY WITH REQUIREMENTS IN ASHRAE 62.1 SECTION 7.2.2 - "AIR BALANCING."
   C. AIRFLOW VALUES ARE TO BE ADJUSTED TO PLUS OR MINUS 5% OF DESIGN FLOW.
   D. SET DAMPERS TO FAIL SAFE POSITION BEFORE TESTING, ADJUSTING, AND BALANCING.
   E. MARK EQUIPMENT AND BALANCING DEVICES, INCLUDING DAMPER-CONTROL POSITIONS, VALVE POSITION INDICATORS, FAN-SPEED-CONTROL LEVERS, AND SIMILAR CONTROLS AND DEVICES, WITH PAINT OR OTHER SUITABLE, PERMANENT IDENTIFICATION MATERIAL TO SHOW FINAL SETTINGS.
   F. ADJUST FANS TO DELIVER DESIGN AIRFLOW WITHIN THE DESIGN FAN SPEED LISTED BY FAN MANUFACTURER. ANY ADJUSTMENTS MUST BE COMMUNICATED WITH CONTROLS VENDOR WHILE T&B IS ON SITE TO CONFIRM ON BMS AND DOCUMENT THE CHANGE ON THE UI.
   G. AFTER TESTING AND BALANCING IS COMPLETE, OPERATE EACH SYSTEM AND RANDOMLY CHECK MEASUREMENTS TO VERIFY THAT THE SYSTEMS ARE OPERATING IN ACCORDING TO THE FINAL TEST AND BALANCE READINGS.

8. OPERATION PRIOR TO ACCEPTANCE: CONTRACTOR MAY OPERATE ANY EQUIPMENT PROVIDED THAT THE OPERATION IS SUPERVISED AND THE CONTRACTOR RETAINS FULL RESPONSIBILITY FOR PROPERLY MAINTAINING EQUIPMENT AND THE FULL MANUFACTURER'S WARRANTY REMAINS UNAFFECTED FROM THE TIME OF OWNER'S POSSESSION.

GENERAL NOTES

A. CONTRACTOR MAY OPERATE ANY EQUIPMENT PROVIDED THAT THE OPERATION IS SUPERVISED AND THE CONTRACTOR RETAINS FULL RESPONSIBILITY FOR PROPERLY MAINTAINING EQUIPMENT AND THE FULL MANUFACTURER'S WARRANTY REMAINS UNAFFECTED FROM THE TIME OF OWNER'S POSSESSION.

B. CONTRACTOR SHALL FURNISH A WRITTEN GUARANTEE COVERING ALL WORKMANSHIP AND MATERIALS FOR A PERIOD OF ONE YEAR FROM DATE OF ACCEPTANCE.

C. CONTRACTOR SHALL PROVIDE POSITIVE MEANS FOR BALANCING EACH INDIVIDUAL AIR CONDITIONING SUPPLY AIR OUTLET, AS PER SCHEDULES & DRAWINGS, SYSTEMS SHALL BE BALANCED AGAINST THE ACTUAL INSTALLED STATIC PRESSURE.

D. BEFORE SUBMITTING A FINAL PROPOSAL, THE CONTRACTOR SHALL EXAMINE THE SITE OF THE PROPOSED WORK TO DETERMINE THE EXISTING CONDITIONS THAT MAY AFFECT HIS WORK AS HE WILL BE RESPONSIBLE FOR ANY ASSUMPTIONS MADE BY HIM IN REGARDS HERETO.

E. ANY APPARATUS, APPLIANCE, MATERIAL, WORK OR INCIDENTAL ACCESSORIES OR MINOR DETAILS NOT SHOWN BUT NECESSARY TO MAKE THE WORK COMPLETE AND PERFECT IN ALL RESPECTS AND READY FOR OPERATION, EVEN IF NOT PARTICULARLY SPECIFIED SHALL BE PROVIDED BY THE CONTRACTOR WITHOUT ADDITIONAL EXPENSE TO THE OWNER.

F. CONTRACTOR SHALL ASSUME RESPONSIBILITY FOR ALL COSTS INCURRED RESULTING FROM SUBSTITUTION OF EQUIPMENT AS WELL AS THE PERFORMANCE AND SPACE REQUIREMENTS OF SUCH EQUIPMENT.

G. ALL TESTS SHALL BE PERFORMED AS REQUIRED DURING THE DIFFERENT STAGES OF WORK AND A FINAL 24 HOUR MINIMUM RUNNING TEST SHALL BE DONE AFTER ALL OTHER TESTS AND BALANCING OPERATIONS HAVE BEEN DONE.

H. TEST AND BALANCE OF ALL EQUIPMENT SHALL BE PERFORMED BY AN INDEPENDENT TEST AND BALANCE COMPANY WITH A MINIMUM OF 5 YEARS EXPERIENCE IN PROJECTS OF THIS SIZE OR LARGER. THREE COPIES OF THE TEST AND BALANCE REPORT SHALL BE SUBMITTED TO THE ENGINEER FOR REVIEW AND APPROVAL.

I. CONTRACT DOCUMENTS FOR THE WORK ARE BASED ON AS-BUILT DRAWINGS PROVIDED BY OWNER. ANY DISCREPANCIES BETWEEN THE WORK DEPICTED IN THE DOCUMENTS AND THE ACTUAL CONDITIONS ENCOUNTERED SHALL BE BROUGHT PROMPTLY TO THE EOR FOR REVIEW AND CLARIFICATION.

J. ANY ADDITIONAL DUCTWORK REQUIRED TO MAKE FINAL EXHAUST DUCTWORK CONNECTIONS SHALL BE METAL PER FMC 501.4. CONSTRUCTION SHALL COMPLY WITH FMC 603. ANY DEFICIENCIES IN EXISTING DUCTWORK CONSTRUCTION SHALL BE REMOVED AND MADE COMPLIANT WITH THE ABOVE LISTED CODE SECTIONS.

K. ALL INDICATED AIR FLOWS ARE THE MINIMUM REQUIRED FOR THE ZONE. IF MORE AIR FLOW IS PROVIDED DISTRIBUTE THE ACCESS AIR PROPORTIONALLY, BY AREA, TO EACH ZONE SUPPLIED BY THE SYSTEM.

L. ALL RUNOUTS TO AIR TERMINALS TO BE SAME SIZE AS AIR TERMINAL CONNECTION UNLESS OTHERWISE NOTED.

M. PROVIDE VOLUME BALANCING DAMPERS FOR ALL DUCT BRANCHES CONNECTED TO AIR TERMINALS. ALL DUCTWORK TO BE SEALED SMACNA CLASS A.

N. T-STATS AND SENSORS TO BE MOUNTED NO MORE THAN 48" ABOVE FINISHED FLOOR, UNLESS OTHERWISE NOTED. PROVIDE 55° F DISCHARGE AIR TEMPERATURE FROM AIR HANDLING UNIT. ALL REQUIRED THERMOSTATS SHALL BE LOCATED AS SHOWN ON DRAWINGS OR WITHIN THE ZONE OF THE UNIT THEY ARE CONTROLLING, UNLESS OTHERWISE NOTED.

O. SUPPORT SUSPENDED MECHANICAL EQUIPMENT WITH ALL-THREAD RODS AND VIBRATION ISOLATORS.

P. PROVIDE GENERAL CONTRACTOR WITH FINAL SIZES AND LOCATIONS OF ALL WALL, FLOOR, AND ROOF PENETRATIONS TO COORDINATE ANY REQUIRED STRUCTURAL FRAMING MEMBERS.

Q. ALL PENETRATIONS OF DUCTWORK, PIPING CONDUITS, AND VENTS THROUGH FIRE-RATED AND SMOKE-RATED BARRIERS SHALL HAVE FIRESTOP AND/OR SMOKE STOP PROTECTION IN ACCORDANCE WITH THE MINIMUM BUILDING CODE.

R. IF AC UNITS MUST BE USED DURING CONSTRUCTION, THEY SHALL HAVE FILTRATION MEDIA WITH A MERV 8 AT EACH RETURN AIR GRILLE. THESE MUST BE REPLACED PRIOR TO OCCUPANCY.

S. HOLD ALL PIPING AND DUCTWORK TIGHT TO STRUCTURE IN ALL AREAS. NOTE THERE ARE HIGH CEILINGS IN MANY AREAS WHICH WILL REQUIRE CLOSE COORDINATION WITH OTHER TRADES DURING SHOP DRAWING PREPARATION AND DURING CONSTRUCTION.

T. SMOKE DETECTORS AND REMOTE ANNUNCIATOR ARE SUPPLIED BY MECHANICAL CONTRACTOR. SMOKE DETECTORS SHALL BE INSTALLED AND INTERLOCKED FOR SHUTDOWN IN ACCORDANCE WITH DIV. 23. SEE SPECIFICATIONS.

U. THREE SETS OF FILTERS ARE TO BE PROVIDED BY CONTRACTOR FOR EACH AC UNIT. ONE SET IS FOR USE DURING CONSTRUCTION, ONE SET TO BE INSTALLED PRIOR TO TEST AND BALANCE, AND A FINAL SET TO BE INSTALLED JUST PRIOR TO OCCUPANCY.

V. ALL MECHANICAL EQUIPMENT APPROACHING END OF USEFUL LIFE (15 YEARS) WILL BE REPLACED WITH LIKE SIZE EQUIPMENT. EFFICIENCIES SHALL BE COMPLIANT WITH LOCAL CODES.

W. CONTRACTOR SHALL BE RESPONSIBLE FOR INSTALLING ALL NECESSARY ACCESS PANELS TO PROVIDE ACCESS TO CONCEALED EQUIPMENT AND BALANCE DAMPERS, REGARDLESS OF WHETHER SAID ACCESS PANELS ARE SHOWN ON THE PLANS.

X. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE HVAC SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

Y. CERTAIN ITEMS SUCH AS RISES AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR CLARITY FOR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS.

Z. IN CORRIDORS WHERE CEILING SPEAKERS AND AIR DIFFUSERS ARE INDICATED BETWEEN THE SAME LIGHT FIXTURES, INSTALL BOTH DEVICES AT THE QUARTER POINTS BETWEEN THE SAME FIXTURE.

AA. ALL DUCTWORK SHALL CLEAR DOORS AND WINDOWS.

BB. UNLESS OTHERWISE NOTED, ALL DUCTWORK IS OVERHEAD, TIGHT TO THE UNDERSIDE OF THE STRUCTURE, WITH SPACE FOR INSULATION IF REQUIRED.

CC. RUNS OF FLEXIBLE DUCT SHALL NOT EXCEED 5 FEET (EDIT MAXIMUM LENGTH OF FLEXIBLE DUCT TO SUIT PROJECT; 5 FEET MAXIMUM RECOMMENDED LENGTH, 8 FEET MAXIMUM LENGTH).

DD. ALL DUCTWORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN DUCTS, INCLUDING DIVIDED DUCTS AND TRANSITIONS AROUND OBSTRUCTIONS, SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.

EE. PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.

FF. PROVIDE ACCESS DOORS IN DUCTWORK FOR OPERATION, ADJUSTMENT, AND MAINTENANCE OF ALL FANS, VALVES, AND MECHANICAL EQUIPMENT.

GG. ALL DUCTS SHALL BE GROUNDED ACROSS FLEXIBLE CONNECTIONS WITH FLEXIBLE COPPER GROUNDING STRAPS. GROUNDING STRAPS SHALL BE BOLTED OR SOLDERED TO BOTH THE EQUIPMENT AND THE DUCT.

HH. SMOKE DETECTORS SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR MOUNTING THE SMOKE DETECTOR IN DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURER'S PRINTED INSTRUCTIONS.

II. TERMINATE GAS VENTS FOR UNIT HEATERS, WATER HEATERS, HIGH PRESSURE PARTS WASHER, HIGH PRESSURE CLEANER, AND OTHER GAS APPLIANCES A MINIMUM OF 3'0" ABOVE ROOF WITH RAIN CAP (EDIT APPLIANCES AND HEIGHT ABOVE ROOF TO MEET CODE AND TO SUIT PROJECT REQUIREMENTS).

JJ. SEE SPECIFICATIONS FOR DUCTWORK GAUGES, BRACING, HANGERS, AND OTHER REQUIREMENTS.

KK. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. DETAILED DESCRIPTIONS ARE PROVIDED IN THE ARCHITECTURAL SPECIFICATIONS.

LL. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, AND DETAILS SHALL BE COORDINATED WITH GENERAL CONTRACTOR.

MM. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, MOUNTING, AND DETAILS SHALL BE COORDINATED WITH OTHER TRADES INVOLVED.

DISCLAIMER NOTE: NOT ALL SPECIFICATIONS OR NOTES ON THIS SHEET MAY APPLY TO THIS SPECIFIC PROJECT. REFER TO PROJECT MECHANICAL PLANS FOR SCOPE OF WORK APPLICABLE TO THIS PROJECT.

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:

FIFTH THIRD BANK

Project Number: 26032
Project Name:

FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:

MECHANICAL NOTES AND SPECIFICATIONS

Sheet Number:

M-001

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HVAC PIPING NOTES

A. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE PIPING SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

B. ELEVATIONS AS SHOWN ON THE DRAWINGS ARE TO THE CENTERLINE OF ALL PRESSURE PIPING AND TO THE INVERT OF ALL GRAVITY PIPING.

C. MAINTAIN A MINIMUM OF 3'6" OF GROUND COVER OVER ALL UNDERGROUND HVAC PIPING.

D. UNLESS OTHERWISE NOTED, ALL CHILLED WATER AND HEATING WATER PIPING SHALL BE 3⁄4 INCH SIZE.

E. PROVIDE AN AIR VENT AT THE HIGH POINT OF EACH DROP IN THE HEATING WATER, CHILLED WATER, AND OTHER CLOSED WATER PIPING SYSTEMS. ALL PIPING SHALL GRADE TO LOW POINTS. PROVIDE HOSE END DRAIN VALVES AT THE BOTTOM OF ALL RISERS AND LOW POINTS.

F. UNLESS OTHERWISE NOTED, ALL PIPING IS OVERHEAD, TIGHT TO UNDERSIDE OF STRUCTURE OR SLAB, WITH SPACE FOR INSULATION IF REQUIRED.

G. INSTALL PIPING SO THAT ALL VALVES, STRAINERS, UNIONS, TRAPS, FLANGES, AND OTHER APPURTENANCES REQUIRING ACCESS ARE ACCESSIBLE.

H. ALL VALVES SHALL BE INSTALLED SO THAT VALVE REMAINS IN SERVICE WHEN EQUIPMENT OR PIPING ON EQUIPMENT SIDE OF VALVE IS REMOVED.

I. ALL BALANCING VALVES AND BUTTERFLY VALVES SHALL BE PROVIDED WITH POSITION INDICATORS AND MAXIMUM ADJUSTABLE STOPS (MEMORY STOPS).

J. PROVIDE CHAINWHEEL OPERATORS FOR ALL VALVES IN EQUIPMENT ROOMS MOUNTED GREATER THAN 7 '-0" ABOVE FLOOR LEVEL; CHAIN SHALL EXTEND TO 7'-0" ABOVE FLOOR LEVEL.

K. ALL VALVES (EXCEPT CONTROL VALVES) AND STRAINERS SHALL BE FULL SIZE OF PIPE BEFORE REDUCING SIZE TO MAKE CONNECTIONS TO EQUIPMENT AND CONTROLS.

L. UNIONS AND/OR FLANGES SHALL BE INSTALLED AT EACH PIECE OF EQUIPMENT, IN BYPASSES, AND IN LONG PIPING RUNS (100 FEET OR MORE) TO PERMIT DISASSEMBLY FOR ALTERATION AND REPAIRS.

M. PITCH STEAM PIPING DOWNWARD IN THE DIRECTION OF FLOW 1-4 INCH IN 10 FEET (1 INCH IN 40 FEET) MINIMUM. PITCH ALL STEAM RETURN LINES DOWNWARD IN THE DIRECTION OF CONDENSATE FLOW 1⁄2 INCH PER 10 FEET (1 INCH IN 20 FEET) MINIMUM. WHERE LENGTH OF BRANCH LINES ARE LESS THAN 8 FEET, PITCH BRANCH LINES TOWARD MAINS 1⁄2 INCH PER FOOT MINIMUM.

N. PITCH UP ALL CONDENSATE RUNOUTS TO RISERS AND EQUIPMENT 1⁄2 INCH PER FOOT. WHERE THIS PITCH CANNOT BE OBTAINED, RUNOUTS OVER 8 FEET IN LENGTH SHALL BE ONE SIZE LARGER THAN NOTED.

O. INSTALL ALL PIPING WITHOUT FORCING OR SPRINGING. ALL PIPING SHALL CLEAR DOORS AND WINDOWS.

P. ALL VALVES SHALL BE ADJUSTED FOR SMOOTH AND EASY OPERATION.

Q. ALL PIPING WORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN PIPING AROUND OBSTRUCTIONS SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.

R. PROVIDE FLEXIBLE CONNECTIONS IN ALL PIPING SYSTEMS CONNECTED TO PUMPS, CHILLERS, COOLING TOWERS, AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION EXCEPT WATER COILS. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AS CLOSE TO THE EQUIPMENT AS POSSIBLE OR AS INDICATED ON THE DRAWINGS.

S. SLOPE REFRIGERANT PIPING ONE PERCENT IN THE DIRECTION OF OIL RETURN. LIQUID LINES MAY BE INSTALLED LEVEL.

T. INSTALL HORIZONTAL REFRIGERANT HOT GAS DISCHARGE PIPING WITH 1⁄2" PER 10 FEET DOWNWARD SLOPE AWAY FROM THE COMPRESSOR.

U. INSTALL HORIZONTAL REFRIGERANT SUCTION LINES WITH 1⁄2" PER 10 FEET DOWNWARD SLOPE TO THE COMPRESSOR, WITH NO LONG TRAPS OR DEAD ENDS WHICH MAY CAUSE OIL TO SEPARATE FROM THE SUCTION GAS AND RETURN TO THE COMPRESSOR IN DAMAGING SLUDGE.

V. PROVIDE LINE SIZE LIQUID INDICATORS IN MAIN LIQUID LINE LEAVING CONDENSER OR RECEIVER. INSTALL MOISTURE-LIQUID INDICATORS IN LIQUID LINES BETWEEN FILTER DRYERS AND THERMOSTATIC EXPANSION VALVES AND IN LIQUID LINE TO RECEIVER.

W. PROVIDE LINE SIZE STRAINER UPSTREAM OF EACH AUTOMATIC VALVE. PROVIDE SHUTOFF VALVE ON EACH SIDE OF STRAINER.

X. PROVIDE PERMANENT FILTER DRYERS IN LOW TEMPERATURE SYSTEMS AND SYSTEMS USING HERMETIC COMPRESSORS. PROVIDE EPLACEABLE CARTRIDGE FILTER DRYERS WITH THREE VALVE BYPASS ASSEMBLY FOR SOLENOID VALVES, ADJACENT TO RECEIVERS.

Y. PROVIDE REFRIGERANT CHARGING VALVE CONNECTIONS IN LIQUID LINE BETWEEN RECEIVER SHUTOFF VALVE AND EXPANSION VALVE.

DISCLAIMER NOTE: NOT ALL SPECIFICATIONS OR NOTES ON THIS SHEET MAY APPLY TO THIS SPECIFIC PROJECT. REFER TO PROJECT MECHANICAL PLANS FOR SCOPE OF WORK APPLICABLE TO THIS PROJECT.

MECHANICAL NOTES

A. CONTRACTOR SHALL FURNISH A WRITTEN GUARANTEE COVERING ALL WORKMANSHIP AND MATERIALS FOR A PERIOD OF ONE YEAR FROM DATE OF ACCEPTANCE.

B. CONTRACTOR SHALL PROVIDE POSITIVE MEANS FOR BALANCING EACH INDIVIDUAL AIR CONDITIONING SUPPLY AIR OUTLET, AS PER SCHEDULES & DRAWINGS, SYSTEMS SHALL BE BALANCED AGAINST THE ACTUAL INSTALLED STATIC PRESSURE.

C. BEFORE SUBMITTING A FINAL PROPOSAL, THE CONTRACTOR SHALL EXAMINE THE SITE OF THE PROPOSED WORK TO DETERMINE THE EXISTING CONDITIONS THAT MAY AFFECT HIS WORK AS HE WILL BE RESPONSIBLE FOR ANY ASSUMPTIONS MADE BY HIM IN REGARDS HERETO.

D. ANY APPARATUS, APPLIANCE, MATERIAL, WORK OR INCIDENTAL ACCESSORIES OR MINOR DETAILS NOT SHOWN BUT NECESSARY TO MAKE THE WORK COMPLETE AND PERFECT IN ALL RESPECTS AND READY FOR OPERATION, EVEN IF NOT PARTICULARLY SPECIFIED SHALL BE PROVIDED BY THE CONTRACTOR WITHOUT ADDITIONAL EXPENSE TO THE OWNER.

E. CONTRACTOR SHALL ASSUME RESPONSIBILITY FOR ALL COSTS INCURRED RESULTING FROM SUBSTITUTION OF EQUIPMENT AS WELL AS THE PERFORMANCE AND SPACE REQUIREMENTS OF SUCH EQUIPMENT.

F. ALL TESTS SHALL BE PERFORMED AS REQUIRED DURING THE DIFFERENT STAGES OF WORK AND A FINAL 24 HOUR MINIMUM RUNNING TEST SHALL BE DONE AFTER ALL OTHER TESTS AND BALANCING OPERATIONS HAVE BEEN DONE.

G. TEST AND BALANCE OF ALL EQUIPMENT SHALL BE PERFORMED BY AN INDEPENDENT TEST AND BALANCE COMPANY WITH A MINIMUM OF 5 YEARS EXPERIENCE IN PROJECTS OF THIS SIZE OR LARGER. THREE COPIES OF THE TEST AND BALANCE REPORT SHALL BE SUBMITTED TO THE ENGINEER FOR REVIEW AND APPROVAL.

H. CONTRACT DOCUMENTS FOR THE WORK ARE BASED ON AS-BUILT DRAWINGS PROVIDED BY OWNER. ANY DISCREPANCIES BETWEEN THE WORK DEPICTED IN THE DOCUMENTS AND THE ACTUAL CONDITIONS ENCOUNTERED SHALL BE BROUGHT PROMPTLY TO THE EOR FOR REVIEW AND CLARIFICATION.

I. ANY ADDITIONAL DUCTWORK REQUIRED TO MAKE FINAL EXHAUST DUCTWORK CONNECTIONS SHALL BE METAL PER FMC 501.4. CONSTRUCTION SHALL COMPLY WITH FMC 603. ANY DEFICIENCIES IN EXISTING DUCTWORK CONSTRUCTION SHALL BE REMOVED AND MADE COMPLIANT WITH THE ABOVE LISTED CODE SECTIONS.

J. ALL INDICATED AIR FLOWS ARE THE MINIMUM REQUIRED FOR THE ZONE. IF MORE AIR FLOW IS PROVIDED DISTRIBUTE THE ACCESS AIR PROPORTIONALLY, BY AREA, TO EACH ZONE SUPPLIED BY THE SYSTEM.

K. ALL RUNOUTS TO AIR TERMINALS TO BE SAME SIZE AS AIR TERMINAL CONNECTION UNLESS OTHERWISE NOTED.

L. PROVIDE VOLUME BALANCING DAMPERS FOR ALL DUCT BRANCHES CONNECTED TO AIR TERMINALS. ALL DUCTWORK TO BE SEALED SMACNA CLASS A.

M. T-STATS AND SENSORS TO BE MOUNTED 44" ABOVE FINISHED FLOOR, UNLESS OTHERWISE NOTED. PROVIDE .55° F DISCHARGE AIR TEMPERATURE FROM AIR HANDLING UNIT. ALL REQUIRED THERMOSTATS SHALL BE LOCATED AS SHOWN ON DRAWINGS OR WITHIN THE ZONE OF THE UNIT THEY ARE CONTROLLING, UNLESS OTHERWISE NOTED.

N. SUPPORT SUSPENDED MECHANICAL EQUIPMENT WITH ALL-THREAD RODS AND VIBRATION ISOLATORS.

O. PROVIDE GENERAL CONTRACTOR WITH FINAL SIZES AND LOCATIONS OF ALL WALL, FLOOR, AND ROOF PENETRATIONS TO COORDINATE ANY REQUIRED STRUCTURAL FRAMING MEMBERS.

P. ALL PENETRATIONS OF DUCTWORK, PIPING CONDUITS, AND VENTS THROUGH FIRE-RATED AND SMOKE-RATED BARRIERS SHALL HAVE FIRESTOP AND/OR SMOKE STOP PROTECTION IN ACCORDANCE WITH THE MINIMUM BUILDING CODE.

Q. IF AC UNITS MUST BE USED DURING CONSTRUCTION, THEY SHALL HAVE FILTRATION MEDIA WITH A MERV 8 AT EACH RETURN AIR GRILLE. THESE MUST BE REPLACED PRIOR TO OCCUPANCY.

R. HOLD ALL PIPING AND DUCTWORK TIGHT TO STRUCTURE IN ALL AREAS. NOTE THERE ARE HIGH CEILINGS IN MANY AREAS WHICH WILL REQUIRE CLOSE COORDINATION WITH OTHER TRADES DURING SHOP DRAWING PREPARATION AND DURING CONSTRUCTION.

S. SMOKE DETECTORS AND REMOTE ANNUNCIATOR ARE SUPPLIED BY MECHANICAL CONTRACTOR. SMOKE DETECTORS SHALL BE INSTALLED AND INTERLOCKED FOR SHUTDOWN IN ACCORDANCE WITH DIV. 23. SEE SPECIFICATIONS.

T. THREE SETS OF FILTERS ARE TO BE PROVIDED BY CONTRACTOR FOR EACH AC UNIT. ONE SET IS FOR USE DURING CONSTRUCTION, ONE SET TO BE INSTALLED PRIOR TO TEST AND BALANCE, AND A FINAL SET TO BE INSTALLED JUST PRIOR TO OCCUPANCY.

U. ALL MECHANICAL EQUIPMENT APPROACHING END OF USEFUL LIFE (15 YEARS) WILL BE REPLACED WITH LIKE SIZE EQUIPMENT. EFFICIENCIES SHALL BE COMPLIANT WITH LOCAL CODES.

V. CONTRACTOR SHALL BE RESPONSIBLE FOR INSTALLING ALL NECESSARY ACCESS PANELS TO PROVIDE ACCESS TO CONCEALED EQUIPMENT AND BALANCE DAMPERS, REGARDLESS OF WHETHER SAID ACCESS PANELS ARE SHOWN ON THE PLANS.

SHEET METAL NOTES

A. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE HVAC SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

B. CERTAIN ITEMS SUCH AS RISES AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR CLARITY FOR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS.

C. IN CORRIDORS WHERE CEILING SPEAKERS AND AIR DIFFUSERS ARE INDICATED BETWEEN THE SAME LIGHT FIXTURES, INSTALL BOTH DEVICES AT THE QUARTER POINTS BETWEEN THE SAME FIXTURE.

D. UNLESS OTHERWISE SHOWN, LOCATE ALL ROOM THERMOSTATS AND HUMIDISTATS 4'-0" (CENTERLINE) ABOVE FINISHED FLOOR. NOTIFY THE ENGINEER OF ANY ROOMS WHERE THE ABOVE LOCATION CANNOT BE MAINTAINED OR WHERE THERE IS A QUESTION ON LOCATION.

E. ALL DUCTWORK SHALL CLEAR DOORS AND WINDOWS.

F. ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERNAL CLEAR DIMENSIONS AND DUCT SIZE SHALL BE INCREASED TO COMPENSATE FOR DUCT LINING THICKNESS.

G. PROVIDE ALL 90 DEGREE SQUARE ELBOWS WITH DOUBLE RADIUS TURNING VANES UNLESS OTHERWISE INDICATED. ELBOWS IN DISHWASHER, KITCHEN, AND LAUNDRY EXHAUST SHALL BE UNVANED SMOOTH RADIUS CONSTRUCTION WITH A RADIUS EQUAL TO 1 1⁄2 TIMES THE WIDTH OF THE DUCT. PROVIDE ACCESS DOORS UPSTREAM OF ALL ELBOWS WITH TURNING VANES.

H. COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, LIGHTING, AND OTHER CEILING ITEMS AND MAKE MINOR DUCT MODIFICATIONS TO SUIT.

I. FIELD ERECTED AND FACTORY ASSEMBLED AIR HANDLING UNIT COILS SHALL BE ARRANGED FOR REMOVAL FROM THE UPSTREAM SIDE WITHOUT DISMANTLING SUPPORTS. PROVIDE GALVANIZED STRUCTURAL STEEL SUPPORTS FOR ALL COILS (EXCEPT LOWEST COIL) IN BANKS OVER TWO COILS HIGH TO PERMIT INDEPENDENT REMOVAL OF ANY COIL.

J. ALL AC UNITS SHALL OPERATE WITHOUT MOISTURE CARRYOVER.

K. LOCATE ALL MECHANICAL EQUIPMENT (SINGLE DUCT, DUAL DUCT, VARIABLE VOLUME, CONSTANT VOLUME AND FAN POWERED BOXES, FAN COIL UNITS, CABINET HEATERS, UNIT HEATERS, UNIT VENTILATORS, COILS, STEAM HUMIDIFIERS, ETC.) FOR UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS AND VALVING.

L. FINNED TUBE RADIATION ENCLOSURES SHALL BE WALL TO WALL UNLESS OTHERWISE INDICATED.

M. PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS (SUPPLY, RETURN, AND EXHAUST) CONNECTED TO AIR HANDLING UNITS, FANS, AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AT THE POINT OF CONNECTION TO THE EQUIPMENT UNLESS OTHERWISE INDICATED.

N. UNLESS OTHERWISE NOTED, ALL DUCTWORK IS OVERHEAD, TIGHT TO THE UNDERSIDE OF THE STRUCTURE, WITH SPACE FOR INSULATION IF REQUIRED.

O. RUNS OF FLEXIBLE DUCT SHALL NOT EXCEED 5 FEET (EDIT MAXIMUM LENGTH OF FLEXIBLE DUCT TO SUIT PROJECT; 5 FEET MAXIMUM RECOMMENDED LENGTH, 8 FEET MAXIMUM LENGTH).

P. ALL DUCTWORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN DUCTS, INCLUDING DIVIDED DUCTS AND TRANSITIONS AROUND OBSTRUCTIONS, SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.

Q. PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.

R. PROVIDE ACCESS DOORS IN DUCTWORK FOR OPERATION, ADJUSTMENT, AND MAINTENANCE OF ALL FANS, VALVES, AND MECHANICAL EQUIPMENT.

S. ALL DUCTS SHALL BE GROUNDED ACROSS FLEXIBLE CONNECTIONS WITH FLEXIBLE COPPER GROUNDING STRAPS. GROUNDING STRAPS SHALL BE BOLTED OR SOLDERED TO BOTH THE EQUIPMENT AND THE DUCT.

T. SMOKE DETECTORS SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR MOUNTING THE SMOKE DETECTOR IN DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURER'S PRINTED INSTRUCTIONS.

U. TERMINATE GAS VENTS FOR UNIT HEATERS, WATER HEATERS, HIGH PRESSURE PARTS WASHER, HIGH PRESSURE CLEANER, AND OTHER GAS APPLIANCES A MINIMUM OF 3'0" ABOVE ROOF WITH RAIN CAP (EDIT APPLIANCES AND HEIGHT ABOVE ROOF TO MEET CODE AND TO SUIT PROJECT REQUIREMENTS).

V. SEE SPECIFICATIONS FOR DUCTWORK GAUGES, BRACING, HANGERS, AND OTHER REQUIREMENTS. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. DETAILED DESCRIPTIONS ARE PROVIDED IN THE ARCHITECTURAL SPECIFICATIONS.

W. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, AND DETAILS SHALL BE COORDINATED WITH GENERAL CONTRACTOR.

X. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, MOUNTING, AND DETAILS SHALL BE COORDINATED WITH OTHER TRADES INVOLVED.

ELECTRICAL REQUIRMENTS FOR MECHANICAL EQUIPMENT

GENERAL REQUIREMENTS

A. DIVISION 23 CONTRACTOR SHALL PROVIDE MECHANICAL EQUIPMENT WITH VOLTAGES AND OTHER ELECTRICAL CHARACTERISTICS AS INDICATED ON THE DRAWINGS AND WITHIN THE SPECIFICATIONS.

B. ALL STARTERS, DISCONNECT SWITCHES, MOTOR CONTROL CENTERS, AND VARIABLE FREQUENCY DRIVES, FOR EQUIPMENT PROVIDED UNDER DIVISION 23, SHALL BE FURNISHED UNDER DIVISION 23 AND INSTALLED UNDER DIVISION 26. FACTORY MOUNTED STARTERS, DISCONNECTS SWITCHES, AND VARIABLE FREQUENCY DRIVES SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23. HVAC CONTRACTOR SHALL PROVIDE STARTERS, DISCONNECT SWITCHES, MOTOR CONTROL CENTERS AND VARIABLE FREQUENCY DRIVES IN ACCORDANCE WITH DIVISION 26.

C. DIVISION 23 SHALL SUBMIT WIRING DIAGRAMS TO THE ARCHITECT/ENGINEER FOR APPROVAL AND PROVIDE APPROVED DIAGRAMS TO THE DIVISION 26 CONTRACTOR SO THAT THE ELECTRICAL WORK MAY BE PROPERLY ACCOMPLISHED.

D. DIVISION 23 CONTRACTOR SHALL FULLY COOPERATE WITH THE OTHER DIVISIONS AND TRADES ON THE PROJECT, AND THEIR MANUFACTURERS, IN PROMPTLY PROVIDING THE INFORMATION REQUIRED FOR PROPER COORDINATION OF MOTOR PROTECTION, CONTROL EQUIPMENT AND WIRING, AND THE OTHER CHARACTERISTICS OF THE EQUIPMENT.

E. IT SHALL BE THE RESPONSIBILITY OF THE DIVISION 23 AND DIVISION 26 CONTRACTORS TO CHECK FOR ADEQUACY OF SUPPLY WIRING, OVERCURRENT PROTECTION, PROPER VOLTAGE, PHASE ROTATION AND FINAL LOCATION OF EQUIPMENT PROVIDED PRIOR TO THE RUNNING OF ANY CONDUIT OR WIRING. COORDINATE WITH DIVISION 26 TO ASSURE PROPER ELECTRICAL SERVICE IS PROVIDED TO EQUIPMENT UNDER DIVISION 23.

F. EQUIPMENT CONNECTIONS SHALL BE MADE THROUGH CONDUIT OR RACEWAYS IN ACCORDANCE WITH DIVISION 26, WITH THE EXCEPTION THAT CONNECTIONS TO MOTORS SHALL BE MADE THROUGH LIQUID TIGHT FLEXIBLE METAL CONDUIT WITH EQUIPMENT GROUNDING CONDUCTOR.

POWER REQUIREMENTS

A. ALL POWER WIRING FOR MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26 TO THE POINT OF FINAL CONNECTION (FROM THE SOURCE TO THE STARTERS, DISCONNECT SWITCHES, MCC'S OR VFD'S AND TO EQUIPMENT, MOTOR, OR OTHER CONNECTION POINT). EXCEPT WHERE SPECIFICALLY INDICATED OTHERWISE, ALL POWER WIRING TO THE POINT OF FINAL CONNECTION FOR EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE ACCOMPLISHED UNDER DIVISION 26.

B. FINAL ELECTRICAL POWER CONNECTIONS TO ALL EQUIPMENT SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26. IN GENERAL, THE POINT OF FINAL CONNECTION SHALL BE THE TERMINAL HOUSING ON THE EQUIPMENT, MOTOR, OR AN INTERNAL JUNCTION BOX ON THE EQUIPMENT OR ITEM. IF NO JUNCTION BOX IS FURNISHED, A JUNCTION BOX SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26. WIRE LEADS F ADEQUATE LENGTH TO ENSURE A PROPER CONNECTION AT THE FINAL LOCATION SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26.

C. WIRING, OVERCURRENT PROTECTION DEVICES, VOLTAGE, PHASE, ROTATION AND FINAL LOCATION OF ALL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE COORDINATED WITH ALL SIMILAR DEVICES AND POWER WIRING URNISHED AND INSTALLED UNDER DIVISION 26. COORDINATION SHALL BE ACCOMPLISHED PRIOR TO THE RUNNING OF ANY CONDUIT OR WIRING.

CONTROL REQUIREMENTS

A. ALL CONTROL WIRING (LINE VOLTAGE AND/OR LOW VOLTAGE) FOR MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23, SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23. WIRING FROM THE POWER SOURCE TO ALL CONTROL PANELS, DDC CONTROL PANELS, STARTERS, OPERATOR WORK STATIONS (PC'S, PRINTERS, MONITORS, AND OTHER WORK STATION EQUIPMENT) AND OTHER CONTROL EQUIPMENT REQUIRED FOR A COMPLETE AND OPERABLE CONTROL SYSTEM SERVING MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23.

B. ALL WIRING FROM CONTROL PANELS TO CONTROL DEVICES FOR MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23. ALL CONTROL WIRE AND CONDUIT SHALL COMPLY WITH THE NATIONAL ELECTRIC CODE AND DIVISION 26 OF THE SPECIFICATIONS. ALL CONTROL WIRING SHALL BE IN CONDUIT.

C. UNLESS OTHERWISE NOTED, ALL TEMPERATURE CONTROL DEVICES, INSTALLED PILOT DEVICES, AND ALL ASSOCIATED PILOT CONTROL EQUIPMENT, FOR ALL EQUIPMENT PROVIDED UNDER DIVISION 23, SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23.

D. ELECTRICAL CONTROL WIRING FOR CONNECTION OF TEMPERATURE CONTROLLERS, PUSH BUTTONS, INTERLOCKS IN MOTOR CONTROLLERS, PNEUMATIC SWITCHES AND LIKE ITEMS IS SPECIFIED UNDER THE CONTROL SECTION(S) IN DIVISION 23 AND INSTALLED BY DIVISION 23.

E. ALL MOTORS, MOUNTS, REMOTE MOUNTED PUSH BUTTON CONTROLS AND ALL SPEED CONTROL SWITCHES FOR MULTI- SPEED MOTORS FOR ALL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23.

HVAC SYMBOLS:

- T — THERMOSTAT CONTROL
- SAT — DUCT MOUNTED SUPPLY AIR TEMPERATURE SENSOR
- RAT — DUCT MOUNTED RETURN AIR TEMPERATURE SENSOR
- AT — AVERAGING TEMPERATURE SENSOR
- HS — HUMIDITY SENSOR
- SD — SMOKE DETECTOR
- C — CARBON DIOXIDE (CO2) SENSOR
- U — 3/4" DOOR UNDERCUT
- G — OPEN VAULT DOOR WITH BARS
- FT — REMOTE FIRE TESTING STATION
- V — VOC SENSOR IN SUPPLY DUCT. EQUAL TO BUILDING AUTOMATION PRODUCTS, INC (BAPI). COORDINATE WITH IOT.
- PC — PARTICLE COUNTER IN RETURN DUCT. EQUAL TO SENSIRION AG SP530. COORDINATE WITH IOT.
- TR — TEMPERATURE AND RELATIVE HUMIDITY DISPLAY EQUAL TO AUTOMATION COMPONENTS, INC. COORDINATE WITH IOT.

HVAC/PIPING SYMBOLS:

- PIPE ELBOW UP
- PIPE ELBOW DOWN
- UNION (FLANGED CONNECTION/JOINT)
- END CAP
- TOP CONNECTION
- BOTTOM CONNECTION
- DIRECTION OF FLOW
- DIRECTION OF PITCH
- D — CONDENSATE DRAIN PIPING
- RL — REFRIGERANT LIQUID PIPING
- RS — REFRIGERANT SUCTION PIPING
- CLEAN OUT PIPING

HVAC LEGEND DOUBLE LINE:

- 12X6 — INTERIOR DUCT DIMENSIONS (W X D)
- DUCT TURNED UP (RETURN)
- DUCT TURNED DOWN (SUPPLY)
- R — RISE IN ELEVATION DIRECTION OF AIRFLOW
- D — DROP IN ELEVATION DIRECTION OF AIRFLOW
- DUCT TURN WITH RADIUS ELBOW
- SQUARE TURN WITH VANE ELBOW
- FLEXIBLE CONNECTION
- DUCT WITH INTERNAL LINING
- TEE-BOOT ENTRY BRANCH WITH MANUAL VOLUME CONTROL DAMPER
- TEE-ROUND ENTRY BRANCH WITH MANUAL VOLUME CONTROL DAMPER
- MOD — MOTOR OPERATED DAMPER
- AP — ACCESS PANELS
- SUPPLY AIR DUCT SECTION
- RETURN AIR DUCT SECTION
- OUTSIDE AIR DUCT SECTION
- EXHAUST AIR DUCT SECTION
- FLEXIBLE AIR DUCT
- RECTANGULAR TO ROUND DUCT TRANSITION
- FD — FIRE DAMPER IN HORIZONTAL DUCT
- FD — FIRE DAMPER IN VERTICAL DUCT
- FD / FSD — COMBINATION FIRE AND SMOKE DAMPER
- DIFFUSER WITH SECTORIZING BAFFLE (EXAMPLE: THREE WAY SUPPLY DIFFUSER)

HVAC EQUIPMENT INSTALLATION

A. INCLUDE ALL CODE AND CONSTRUCTION RELATED PERMITS.

B. INSTALLATION TO MEET ALL FEDERAL AND LOCAL CODE REQUIREMENTS.

C. RTU CONDENSATE LINES SHALL BE EXTENDED TO ROOF DRAINS, SCUPPERS, OR GUTTERS AND SHALL HAVE A CORRECT TRAP DESIGN.

D. RTUS SHALL BE PROPERLY SEATED AND SEALED ON THE ROOF CURB.

E. THERMOSTAT WIRES SHALL NOT BE SPLICED OUTSIDE OF THE HVAC UNIT. SHALL BE INSTALLED IN CONDUIT SEPARATELY FROM HIGH VOLTAGE AND SHALL NOTE BE INSIDE THE DUCTWORK.

F. FOLLOW OEM INSTRUCTIONS FOR STARTUP AND OPERATIONAL COMMISSIONING.

G. ALL ROOF PENETRATIONS SHALL BE SEALED WATER TIGHT.

H. MECHANICAL CONTRACTOR AND HVAC SYSTEM SUPPLIER SHALL CONFIRM WITH CONTROLS VENDOR TO ENSURE FULL COMPATIBILITY FOR ALL HVAC EQUIPMENT PRIOR TO ORDERING/PURCHASE.

I. MC, EC, AND CONTROL VENDOR TO ATTEND A CONTROLS INSTALLATION / DESIGN MEETING.

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:

FIFTH THIRD BANK

Project Number: 26032
Project Name:

FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:

MECHANICAL NOTES AND LEGEND

Sheet Number:

M-002

6/30/2026 4:54:50 PM

MINIMUM DUCT INSULATION R-VALUES:

LOCATION (1)	SUPPLY DUCT	RETURN DUCT
EXTERIOR OF BUILDING	R-8	R-12
VENTILATED ATTIC	R-8	R-8
UNVENTED ATTIC ABOVE INSULATED CEILING	R-6	R-6
UNVENTED ATTIC WITH ROOF INSULATION	R-8	R-12
UNCONDITIONED SPACES (2)	R-6	R-6
INDIRECTLY CONDITIONED SPACES (3)	R-6	R-6
CONDITIONED SPACES	NONE	NONE

NOTES:
(1) INCLUDES RETURN AIR PLENUMS WITH OR WITHOUT EXPOSED ROOFS ABOVE.
(2) INCLUDES CRAWL SPACES, BOTH VENTILATED AND NON-VENTILATED.
(3) THIS TABLE APPLIES TO COMMERCIAL HVAC SYSTEMS ONLY AND INCLUDES BOTH SUPPLY AND RETURN DUCTS.
(4) MINIMUM INSULATION VALUES ARE BASED ON IECC C403 AND ASHRAE 90.1, AS ADOPTED BY THE AUTHORITY HAVING JURISDICTION.

EQUIPMENT ROOF STAND HEIGHTS:

MOUNTED MECHANICAL UNITS					
WIDTH OF MECHANICAL UNIT (IN)	< 24	25-36	47-48	49-60	> 60
MINMUM CLEARANCE ABOVE SURFACE	14	18	24	30	48

NOTES:
ROOF MOUNTED MECHANICAL UNITS SHALL BE MOUNTED ON CURBS RAISED A MINIMUM OF 8 INCHES ABOVE THE ROOF SURFACE OR WHERE ROOFING MATERIALS EXTEND BENEATH THE UNIT, ON RAISED EQUIPMENT SUPPORTS PROVIDING A MINIMUM CLEARANCE HEIGHT IN ACCORDANCE WITH TABLE THE TABLE PROVIDED.

MECHANICAL CONTROLS & ACCESSORIES

GENERAL CONTROLS REQUIREMENTS (ALL EQUIPMENT)

MECHANICAL CONTRACTOR(MC) SHALL PROVIDE TEMPORARY DIGITAL THERMOSTATS AS REQUIRED FOR EQUIPMENT START-UP AND INITIAL OPERATION. IOT SHALL FURNISH AND INSTALL ALL PERMANENT KMC CONTROLS AND COMPLETE BMS INTEGRATION AFTER EQUIPMENT START-UP AND PRIOR TO FINAL COMMISIONING.

MOUNT ALL THERMOSTATS AND SENSORS AT 44" AFF TO CENTER LINE OF VERTICAL DEVICE UNLESS NOTED OTHERWISE. COORDINATE FINAL DEVICE COLOR AND FINISH WITH ARCHITECT.

MC SHALL CORDINATE ALL REQUIRED CONTROL WIRING, POWER, INTERFACES, AND ACCESSORIES NECESSARY FOR FULLY FUNCTIONAL AND COMMISSIONED SYSTEM OPERATION.

RTU CONTROLS (CARRIER UNITS)

PERMANENT THERMOSTAT: KMC FLEXSTAT BAC-120063CW
(PURCHASED AND INSTALLED BY IOT)

MONITORING SENSOR: KMC APPROVED SENSORS
(PURCHASED AND INSTALLED BY IOT)

RTU'S TO USE TEMPORARY THERMOSTATS UNTIL THE PERMANENT KMC BRANDED FLEXSTATS WITH BMS INTEGRATION ARE INSTALLED AND COMMISSIONED. UPON CHANGEOVER, TEMPORARY THERMOSTATS SHALL BE REMOVED AND RETURNED TO MC.

RTU ACCESSORIES NOTE:
MECHANICAL CONTRACTOR TO PROVIDE ROOFTOP UNIT WITH THE FOLLOWING ACCESSORIES:

FIELD INSTALLED

1. WALL MOUNTED CO2 SENSOR (PART NO: 33ZCSPTC02-01)
2. RETURN ENTHALPY SENSOR KIT (PART NO: CRENTSEN001A00)
3. OUTDOOR ENTHALPY SENSOR KIT (PART NO: CRENTSEN001A00)
4. WIFI STICK FOR ECONOMIZERS (PART NO: CRWFTCK001A00)

FACTORY INSTALLED

5. ECONOMIZER CONTROLLER (SIEMENS POL224)
6. ACTUATOR (PART NO: HF23BL005)
7. POWER HARNESS (PART NO: 48TC005897)
8. SUPPLY AIR SENSOR (PART NO: HH79NZ039)

FCU THERMOSTAT NOTE:
MECHANICAL CONTRACTOR TO PROVIDE AND INSTALL THERMOSTAT WITH INTERFACE. IOT WILL SUPPLY KMC CONTROLS AND COMPLETE BMS INTEGRATION AFTER EQUIPMENT STARTUP.

THERMOSTAT INTERFACE ADAPTERS (BY MC AS REQUIRED)

- MITSUBISHI / TRANE THERMOSTAT INTERFACE: PAC-US445CN-1
- CARRIER THERMOSTAT INTERFACE: KSAIC0301230
  (EC MAY BE REQUIRED FOR HIGH VOLTAGE WIRING)

SPLIT SYSTEMS SHALL UTILIZE KMC FLEXSTAT WITH BMS INTEGRATION. MINI-SPLITS SHALL UTILIZE OEM-APPROVED THERMOSTAT INTERFACE ADAPTER WITH KMC FLEXSTAT FOR BMS INTEGRATION.

PACKAGED ROOFTOP HEAT PUMP AIR HANDLING UNIT SCHEDULE: GC PROVIDED

MARK	AREA SERVED	MFG.	MODEL	ROOM TEMP	NOMINAL SIZE	SUPPLY AIR FLOW (CFM)	MINIMUM OUTDOOR AIR FLOW (CFM)	FILTER TYPE	EXT S.P. (IN W.C.)	FAN MOTOR		COOLING CAPACITY				HEATING CAPACITY			ELECTRICAL				WEIGHT (LBS)	SEER
										RPM	MOTOR BPH	TOTAL CAPACITY	SENSIBLE CAPACITY	ENTERING AIR (° FDB/°FWB)	MEDIA	HEAT SOURCE	STAGES	MBH	VOLT	PH	HZ	MCA/MOCP (W/ HEAT)		
RTU-1	ENTRY/MEETING	CARRIER	48FEDA05J3M5-8W4D0	75° F	4.0	1,600	--	MERV 13	1.0	--	0.91	52.33 MBH	37.66 MBH	80/67	R-454B	NATURAL GAS	1	67	208V	3	60	24/30	797	14.0
RTU-2	SEATING/LOUNGE	CARRIER	48FEDA05J3M5-8W4D0	75° F	4.0	1600	--	MERV 13	1.0	--	0.91	52.33 MBH	37.66 MBH	80/67	R-454B	NATURAL GAS	1	67	208V	3	60	24/30	797	14.0

SCHEDULE NOTES:
ACCEPTABLE MANUFACTURERS INCLUDE CARRIER (BOD), TRANE, AND DAIKIN GIVEN COMPARABLE MODEL IS SELECTED. FIELD INSTALLED ACCESSORIES ARE ALSO ACCEPTABLE. CONTACT EMAIL FOR CARRIER REPRESENTATIVE: NATIONALACCOUNTS@CARRIER.COM.

1. PROVIDE 18" FACTORY INSULATED ROOF CURB. ROOF CURB TO BE PITCHED 1/4" PER FOOT ACCORDING TO ROOF SLOPE. SHEET METAL CONTRACTOR SHALL PROVIDE EXTENDED ROOF CURB FOR ROUTING OF DUCTWORK WITHIN CURB TO A SINGLE ROOF OPENING, CONFIRM WITH STRUCTURAL THAT DISCHARGE DUCT OPENING IN THE BOTTOM OF THE UNIT FITS BETWEEN THE JOISTS.
2. PROVIDE RTU WITH VERTICAL SUPPLY AND RETURN AIR CONNECTIONS AND DRAW THRU CONFIGURATION WITH STAINLESS STEEL DRAIN PAN.
3. INCLUDE UNIT WITH THE FOLLOWING; REFERENCE ENTHALPY CONTROLLED ECONOMIZER WITH LOW LEAKAGE DAMPERS, BAROMETRIC RELIEF DAMPER, AND OUTSIDE AIR SYSTEM INTAKE HOOD WITH AUTOMATIC DAMPER. EVAPORATOR FAN TO BE DIRECT DRIVE.
4. FURNISH FUSED DISCONNECT FOR INSTALLATION BY ELECTRICAL CONTRACTOR. PROVIDE THROUGH-THE-BASE ELECTRICAL DISCONNECT SWITCH WITH UNPOWERED CONVENIENCE OUTLET ON UNIT. INCLUDE UNIT AUTO RESET PHASE MONITOR.
5. PROVIDE TWO (2) SETS OF 2" MERV 13 CARTRIDGE FILTERS WITH EACH UNIT.
6. PROVIDE FACTORY SEACOAST SPRAY PROTECTION ON UNIT FOR ANY PROJECT LOCATION WITHIN 20 MILES OF A COASTLINE.
7. PROVIDE FACTORY INSTALLED BURGLAR BARS, CONDENSER COIL GUARD, AND EXTERNAL HAIL GUARDS.
8. PROVIDE WITH HURRICANE TIE DOWNS AND REFER TO ENGINEERED WIND LOAD MITIGATION CALCULATIONS.
9. PROVIDE UNITS WITH COIL CONFIGURATION HEAT THEN COOL, COIL LEAVING AIR TEMPERATURES TO BE NO LESS THAN 55°F. UNIT SELECTION SHALL BE BASED ON OUTSIDE AIR TEMPERATURE OF 110°F IN CLIMATE ZONES 1A-2A, ALL OTHER ZONES TO HAVE NET COOLING CAPACITY OF 95°F AMBIENT. HEAD PRESSURE CONTROL TO 0°F AMBIENT.
10. UNIT MOUNTED SIEMENS POL224 ECONOMIZER CONTROLLER. TEMPORARY THERMOSTATS SHALL BE DIGITAL THERMOSTATS FOR START UP THEN REPLACED BY IOT WITH FIFTH THIRD BMS KMC FLEXSTAT. TEMPORARY THERMOSTATS SHALL BE RETURNED TO MC UPON COMPLETION OF THE CONTROLS INSTALLATION. MONITORING SENSORS SHALL BE PROVIDED BY IOT AND INSTALLED BY MC.
11. ALL HVAC SYSTEM CONTROLS SHALL BE SELECTED TO BE COMPATIBLE WITH BMS CONTROL SYSTEM. MECHANICAL CONTRACTOR AND HVAC SYSTEM SUPPLIER SHALL CONFIRM WITH CONTROLS VENDOR TO ENSURE FULL COMPATIBILITY FOR ALL HVAC EQUIPMENT PRIOR TO ORDERING/PURCHASE.
12. PROVIDE DUCT SMOKE DETECTORS IN RETURN AND SUPPLY MAIN DUCT FOR ANY UNITS PROVIDING AT/OVER 2,000CFM OF AIRFLOW.
13. DEMAND CONTROL VENTILATION (DCV) WITH SPACE CO2 SENSOR FROM HVAC EQUIPMENT MANUFACTURER.
14. UNIT SHALL BE ASHRAE 90.1 COMPLIANT.
15. RTU'S TO BE PURCHASED WITH THE ELECTRO-MECHANICAL CONTROLS AND ULTRA LOW LEAK ECONOMIZER WITH BAROMETRIC RELIEF. REFER TO RTU ACCESSORIES NOTE FOR ADDITIONAL INFORMATION.
16. PROVIDE 2" DOUBLE WALL CONSTRUCTION FOR R-13 THERMAL RESISTANCE INCLUDING THE BOTTOM PANEL WITH HINGED ACCESS DOORS.
17. RTU TO COME WITH CONDENSATE OVERFLOW SWITCH - THIS SENSOR AND RELATED CONTROLLER MONITORS THE CONDENSATE LEVEL IN THE DRAIN PAN AND SHUTS DOWN COMPRESSION OPERATION WHEN OVERFLOW CONDITIONS OCCUR. IT INCLUDES:
    - INDICATOR LIGHT - SOLID RED (MORE THAN 10 SECONDS ON WATER CONTACT - COMPRESSORS DISABLED), BLINKING RED (SENSOR DISCONNECTED)
    - 10-SECOND DELAY TO BREAK - ELIMINATES NUISANCE TRIPS FROM SPLASHING OR WAVES IN PAN (SENSOR NEEDS 10 SECONDS OF CONSTANT WATER CONTACT BEFORE TRIPPING)
    - DISABLES THE COMPRESSORS OPERATION WHEN CONDENSATE PLUG IS DETECTED, BUT STILL ALLOWS FANS TO RUN FOR ECONOMIZER.
18. MECHANICAL CONTRACTOR TO NOTIFY IOT IF RTU UNITS ARE NOT EQUIPPED WITH SYSTEM VU CONTROLS. CONTACT GERRIANNE FAULHABER: GERRIANNE@IOFTEAM.COM, 224-213-6908.

MINI-SPLIT SYSTEM AIR CONDITIONER SCHEDULE: GC PROVIDED

MARK		SYSTEM ZONE	MFG.	INDOOR UNIT			OUTDOOR UNIT			COOLING				HEATING			ELECTRICAL				PHYSICAL	
INDOOR UNIT	OUTDOOR UNIT			MODEL NUMBER	MOUNTING	FAN AIRLOW	MODEL NUMBER	SYSTEM TYPE	REFR. TYPE	RATED CAPACITY (BTU/H)	MINIMAL CAPACITY (BTU/H)	SEER	EER	RATED CAPACITY (BTU/H)	MINIMAL CAPACITY (BTU/H)	HSPF	VOLT/PH/HZ	UNIT MCA		MOCP	INDOOR UNIT WEIGHT	OUTDOOR UNIT WEIGHT
																		INDOOR	OUTDOOR			
FCU-1	HP-1	ELECTRICAL / IT ROOM	MITSUBISHI	PKA-A12LNL	WALL MOUNT	385 CFM	PUZ-A12NL	HEAT PUMP	R-454B	12,200	4,300	21.1	14.8	18,000	4,300	10.1	208 / 230V, 1-PHASE, 60HZ	1 A	16 A	27 A	28 LBS	99.21 LBS

SCHEDULE NOTES:
NO ALTERNATIVE MANUFACTURERS OR MODEL NUMBERS ARE ACCEPTABLE. FIFTH THIRD BANK HAS A NATIONAL ACCOUNT AGREEMENT WITH THE TRANE COMPANY. FOR COMPLETE INFORMATION ON THE FOLLOWING EQUIPMENT PACKAGE PLEASE CONTACT TRANE NATIONAL ACCOUNTS: DANIEL SWEGLE AT DANIEL.SWEGLE@TRANE.COM / 636-875-6616.

1. INDOOR UNITS RECEIVE POWER FROM OUTDOOR UNITS THROUGH FIELD-SUPPLIED INTERCONNECTED WIRING. COORDINATE FINAL UNIT INSTALLATION LOCATION WITH STRUCTURAL CONDITIONS.
2. PROVIDE GRAVITY CONDENSATE DRAIN WHERE 1/4" PER FOOT SLOPE IS ACHIEVABLE. WHERE NOT ACHIEVABLE FOR WALL-MOUNTED UNITS, PROVIDE LITTLE GIANT VCMA-15ULST CONDENSATE PUMP WIRED FOR HIGH-WATER SHUTDOWN. DUCTED PEAD UNITS SHALL USE FACTORY CONDENSATE LIFT AND INTERNAL SAFETY; NO ADDITIONAL ACCESSORIES REQUIRED UNLESS LIFT EXCEEDS 27".
3. ZONES FED BY THIS SYSTEM ARE NON-OCCUPIABLE OR ARE NATURALLY VENTILATED.
4. PROVIDE WITH MITSUBISHI PAC-US445CN-1 HARD WIRED BACK-LIT MA REMOTE CONTROLLER. COMPONENTS TO HAVE BACNET INTERFACE AND INTERLOCK WITH BMS, PROVIDE CONTROL WIRING DIAGRAM FOR ENGINEER REVIEW.
5. PROVIDE FACTORY SEACOAST SPRAY PROTECTION ON UNIT FOR ANY PROJECT LOCATIONS WITHIN 20 MILES OF A COASTLINE.
6. REFRIGERANT CIRCUIT ACCESS PORTS LOCATED OUTDOORS SHALL BE FITTED WITH LOCKING TYPE TAMPER RESISTANT CAPS.
7. MOUNT OUTDOOR UNIT ON MINIMUM 18" HIGH NONISOLATED WIND AND SEISMIC RESTRAINT RAILS.BASIS OF DESIGN; VMC GROUP MODEL #R-7000.
8. HANG INDOOR WALL MOUNTED UNIT TO SECURE WALL STRUCTURE PER MANUFACTERE RECOMMENDATIONS (REFER TO INSTALLATION DETAIL).
9. PROVIDE FULL PORT BALL VALVES & SCHRADER VALVES WITH FLARED CONNECTIONS ON INDOOR WALL UNIT.
10. PROVIDE DRAIN PAN WITH UL#508 APPROVED WATER DETECTION SENSOR FOR UNIT SHUTDOWN. WIRE CONTROL CIRCUIT THROUGH NC CONTACT.
11. COOLING CAPACITY RATED AT 95°F AND HEATING CAPACITY RATED AT 17°F.
12. 100% HEATING CAPACITY AT 5°F AND HEATING PERFORMANCE, DOWN TO -13°F WITH AVERAGE OF 80% CAPACITY.
13. ELECTRICAL CONTRACTOR TO INCLUDE AND INSTALL 3-POLE DISCONNECT SWITCH.
14. MECHANICAL CONTRACTOR TO NOTIFY IOT IF DIFFERENT SPLIT SYSTEM IS PROVIDED. CONTACT GERRIANNE FAULHABER: GERRIANNE@IOFTEAM.COM, 224-213-6908.

DUCT SIZING GUIDE

SUPPLY DUCT (0.08SP)				RETURN DUCT (0.05SP)			
ROUND DUCT		RECTANGULAR DUCT		ROUND DUCT		RECTANGULAR DUCT	
CFM RANGE	SIZE (IN)	CFM RANGE	SIZE (IN)	CFM RANGE	SIZE (IN)	CFM RANGE	SIZE (IN)
1-100	6	1-125	6X6	1-75	6	1-100	6X6
101-210	8	126-260	8X8	76-160	8	101-200	8X8
211-395	10	261-470	10X10	161-290	10	201-365	10X10
396-625	12	471-760	12X12	291-470	12	366-595	12X12
626-910	14	761-1,145	14X14	471-710	14	596-895	14X14
911-1,300	16	1,146-1,635	16X16	711-1,010	16	896-1,275	16X16
1,301-1,800	18	1,636-2,235	18X18	1,011-1,375	18	1,276-1,745	18X18
1,801-2,325	20	2,236-2,955	20X20	1,376-1,815	20	1,746-2,310	20X20

NOTES:
1. FOR RECOMMENDED SQUARE SIZE DUCT AN APPROXIMATE EQUIVALENT RECTANGULAR CROSS SECTIONAL AREA IS ALSO ACCEPTABLE FOR ALTERNATIVE RECTANGULAR DUCT DIMENSIONS. DO NOT EXCEED A 3:1 DIMENSION RATIO.
2. ROUND AND SQUARE DUCT DIMENSIONS HAVE BEEN SIZED BASED ON 0.08 INCHES OF FRICTION HEAD LOSS FOR SUPPLY DUCT AND 0.05 FOR RETURN DUCT.

AIR DISTRIBUTION SCHEDULE

MARK	NECK SIZE	MAX CFM	DESCRIPTION
S (CFM)	6"ø	0-90	FACE SIZE SHALL BE 12X12 (SEE BELOW FOR REMAINING NOTES)
	6"ø	0-100	ARCHITECTURAL SQUARE PANEL CEILING SUPPLY AIR DIFFUSER - STEEL CONSTRUCTION, MODULE SIZE 24X24, CURVED BACK-PAN DESIGN AND ROUND DUCT CONNECTIONS, PROVIDE ROUND RUN-OUTS AS INDICATED BELOW; TITUS D-OMNI-1 (BORDER TYPE 3; LAY-IN FULL FACE FLAT T-BAR CEILING GRID) COORDINATE CEILING TYPE WITH ARCHITECTURAL RCP. COORDINATE LOCATIONS WITH LIGHTING DO NOT SCALE FROM MECHANICAL DRAWINGS
	8"ø	101-210	
	10"ø	211-330	
	12"ø	331-480	
	14"ø	481-650	
LSD (X)	(X) NUMBER OF 1" SLOTS		MODULINEAR SLOT SUPPLY CEILING DIFFUSER WITH 1" SLOT, 4'-0" LONG WITH INSULATED PLENUM BOX, END BORDERS, AND FACE OPERATED REMOTE DAMPER AT PLENUM INLET. COORDINATE CEILING TYPE WITH ARCHITECTURAL RCP. DESIGN MAKE; TITUS MLTI-39, PROVIDE WITH INSULATED PLENUM BOX.
LSR (X)	(X) NUMBER OF 1" SLOTS		MODULINEAR SLOT RETURN CEILING DIFFUSER, 4'-0" LONG, WITH NO PATTERN CONTROLLER, FOR RETURN AIR DUTY. COORDINATE CEILING TYPE WITH ARCHITECTURAL RCP. DESIGN MAKE; TITUS MLTI-39, PROVIDE WITH INSULATED PLENUM BOX.
LD (X)	(X) NUMBER OF 1" SLOTS		MODULINEAR SLOT SUPPLY CEILING DIFFUSER, 2'-0" LONG WITH INSULATED PLENUM BOX, END BORDERS, AND FACE OPERATED REMOTE DAMPER AT PLENUM INLET. COORDINATE CEILING TYPE WITH ARCHITECTURAL RCP. DESIGN MAKE; TITUS MLTI-39, PROVIDE WITH INSULATED PLENUM BOX.
R (CFM) TR (CFM)	10" X10"	0-400	EGGCRATE RETURN AIR GRILLE - CORE OF 1/2"X1/2"X1" ALUMINUM GRID. TYPICAL PANEL SIZE; 12X12, 24X12, 24X24, AND 24X8. ALL CEILING RETURN, EXHAUST, TRANSFER GRILLES TO BE TITUS 50F (BORDER TYPE 3; LAY-IN FULL FACE FLAT T-BAR CEILING GRID) COORDINATE CEILING TYPE WITH ARCHITECTURAL RCP. SIZE AS INDICATED IN TABLE (PROVIDE SQUARE TO ROUND TRANSITION AS NEEDED). COORDINATE LOCATIONS WITH LIGHTING DO NOT SCALE FROM MECHANICAL DRAWINGS
	22" X 10"	401-800	
	22" X 22"	801-2,000	
	22" X 46"	2,001-4,000	

NOTES:
1. FURNISH WITH STANDARD WHITE BAKED ENAMEL FINISH (#26 WHITE). SUBMIT COLOR SAMPLE FOR ALL AIR DEVICES TO ARCHITECT BEFORE ORDERING.
2. FURNISH "RAPID-MOUNT" FRAMES FOR S-1 & R-1 DEVICES LOCATED IN DRYWALL CEILINGS. COORDINATE MOUNTING FRAME WITH CEILING/WALL TYPE.
3. PROVIDE 120/240 VAC TRANSFORMER POWER MODULE, CONTROLLER/THERMOSTAT AND PRESSURE RELIEF RING FOR ALL VAV DIFFUSERS.
4. SIZE FOR MAXIMUM NOISE CRITERION <30 FOR DIFFUSERS, REGISTERS, AND GRILLES.
5. PROVIDE DIRECTIONAL BLOW CLIPS WHERE REQUIRED OR DIRECTIONAL NECK BAFFLES WHERE INDICATED FOR BLOCKED AIRFLOW.
6. PROVIDE REMOTE CABLE OPERATED VOLUME DAMPERS LOCATED IN INACCESSIBLE CEILING/WALL CONSTRUCTION.

* BASIS OF DESIGN PRODUCTS ARE INDICATED FOR REFERENCE ONLY. COMPARABLE PRODUCTS WILL BE ACCEPTED PROVIDED COMPLIANCE WITH TECHNICAL SPECIFICATION REQUIREMENTS. ACCEPTABLE MANUFACTURERS SHALL BE PRICE, TITUS, ANEMOSTAT AND KRUEGER.

EXHAUST FAN SCHEDULE: GC PROVIDED

MARK	AREA SERVED	PRODUCT		MOUNTING TYPE	CFM	EXT S.P. (IN W.C.)	MOTOR DATA				NOTES
		MANUFACTURER	MODEL				RPM	TYPE	VOLT/PH/HZ	WATTS	
EF-1	JANITOR'S CLOSET	GREENHECK	SP-B90	CEILING	55	0.33	889	FSC	120/1/60	17 W	(1)(2)(4)(5)(6)
EF-2	MEN'S RESTROOM	GREENHECK	SP-B110	CEILING	80	0.33	650	FSC	120/1/60	30 W	(1)(2)(3)(4)(5)(6)
EF-3	WOMEN'S RESTROOM	GREENHECK	SP-B110	CEILING	80	0.33	650	FSC	120/1/60	30 W	(1)(2)(3)(4)(5)(6)

NOTES:
(1) ACCEPTABLE MANUFACTURERS: LOREN COOK, GREENHECK, OR CAPTIVEAIRE AND MUST MEET EQUIVALENT CHARACTERISTICS.
(2) PROVIDE FANS WITH FSC SPEED CONTROLLER (5A-120V), LOCATE WITHIN ACCESSIBLE LOCATION.
(3) EXHAUST FAN SHALL BE INTERLOCKED WITH LOCAL LIGHTING THAT TURN ON DURING ROOM SENSOR OCCUPIED MODE (COORDINATE WITH EC).
(4) PROVIDE FANS WITH THE FOLLOWING: TYPICAL DISCONNECT SWITCH, INTEGRAL GRAVITY BACKDRAFT DAMPER, AND HANGING ISOLATOR KIT.
(5) PURCHASE FANS WITH ALUMINUM CEILING GRILLE IF STANDARD IS PLASTIC.
(6) PROVIDE MOUNTING FRAME FOR GYPSUM BOARD CEILING.

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:
FIFTH THIRD BANK

Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
MECHANICAL SCHEDULES

Sheet Number:
M-101

6/30/2026 4:54:52 PM

VENTILATION CALCULATIONS: RTU-1

1. Summary

Ventilation Sizing Method	**ASHRAE Standard 62.1-2019**	
Design Condition	**Cooling operation**	
Occupant Diversity (D)	**1.000**	
Uncorrected Outdoor Air Intake (Vou)	**165**	CFM
System Ventilation Efficiency (Ev)	**0.887**	
Outdoor Air Intake (Vot)	**186**	CFM

2. Space Ventilation Analysis

	Supply Air (CFM)	Space Floor Area (sqft)	Area Outdoor Air Rate (CFM/sqft)	Time Averaged Occupancy (Occupants)	People Outdoor Air Rate (CFM/person)	Air Distribution Effectiveness	Space Outdoor Air (CFM)	Breathing Zone Outdoor Air (CFM)	Space Ventilation Efficiency
Zone Name / Space Name	(Vpz)	(Az)	(Ra)	(Pz)	(Rp)	(Ez)	(Voz)	(Vbz)	(Evz)
Zone 1									
100 Weather Vestibule	458	90.1	0.06	2.0	5.00	1.00	**15**	*15*	1.070
101 Entry	168	114.6	0.06	2.0	7.50	1.00	**22**	*22*	0.973
103 Gathering Area	218	243.6	0.06	4.0	7.50	1.00	**45**	*45*	0.898
104 Booth	257	88.8	0.06	3.0	5.00	1.00	**20**	*20*	1.024
105 Meeting Room	208	119.5	0.06	3.0	5.00	1.00	**22**	*22*	0.996
112 Women's	60	67.3	0.12	1.0	5.00	1.00	**13**	*13*	0.887
113 Men's	66	67.4	0.12	1.0	5.00	1.00	**13**	*13*	0.906
114 Phone Room	61	51.2	0.06	1.0	5.00	1.00	**8**	*8*	0.972
116 Vestibule	104	108.5	0.06	2.0	0.00	1.00	**7**	*7*	1.040
Totals	**1600**						**165**		**0.887**

VENTILATION CALCULATIONS: RTU-2

1. Summary

Ventilation Sizing Method	**ASHRAE Standard 62.1-2019**	
Design Condition	**Cooling operation**	
Occupant Diversity (D)	**1.000**	
Uncorrected Outdoor Air Intake (Vou)	**140**	CFM
System Ventilation Efficiency (Ev)	**0.916**	
Outdoor Air Intake (Vot)	**153**	CFM

2. Space Ventilation Analysis

	Supply Air (CFM)	Space Floor Area (sqft)	Area Outdoor Air Rate (CFM/sqft)	Time Averaged Occupancy (Occupants)	People Outdoor Air Rate (CFM/person)	Air Distribution Effectiveness	Space Outdoor Air (CFM)	Breathing Zone Outdoor Air (CFM)	Space Ventilation Efficiency
Zone Name / Space Name	(Vpz)	(Az)	(Ra)	(Pz)	(Rp)	(Ez)	(Voz)	(Vbz)	(Evz)
Zone 2									
102 Relaxed Seated Area	742	305.5	0.06	3.0	7.50	1.00	**41**	*41*	1.033
107 Library	139	146.7	0.06	2.0	7.50	1.00	**24**	*24*	0.916
108 Transaction Bar	276	179.8	0.06	2.0	7.50	1.00	**26**	*26*	0.994
110 Employee Lounge	328	189.9	0.12	2.0	5.00	1.00	**33**	*33*	0.988
111 Work Room	114	114.9	0.06	2.0	5.00	1.00	**17**	*17*	0.939
Totals	**1600**						**140**		**0.916**

COMCHECK REPORT

COMcheck Software Version 4.1.5.5

Mechanical Compliance Certificate

Project Information

Energy Code:	2015 IECC
Project Title:	
Location:	Warner Robins, Georgia
Climate Zone:	3a
Project Type:	New Construction
Construction Site:	Owner/Agent: Designer/Contractor:
GA	

Additional Efficiency Package(s)
Credits: 12 Required: 0.0 Proposed

Mechanical Systems List

Quantity System Type & Description

1 RTU-1 (Single Zone)
Heating: 1 each - Other, Electric, Capacity = 60 kBtuh
No minimum efficiency requirement applies
Cooling: 1 each - Single Package DX Unit, Capacity = 49 kBtuh, Air-Cooled Condenser, Air Economizer
Proposed Efficiency = 17.00 SEER, Required Efficiency: 14.00 SEER
Fan System: None

1 RTU-2 (Single Zone)
Heating: 1 each - Other, Electric, Capacity = 60 kBtuh
No minimum efficiency requirement applies
Cooling: 1 each - Single Package DX Unit, Capacity = 49 kBtuh, Air-Cooled Condenser, Air Economizer
Proposed Efficiency = 17.00 SEER, Required Efficiency: 14.00 SEER
Fan System: None

1 AHU-1/DU-1 (Single Zone)
Split System Heat Pump
Heating Mode: Capacity = 32 kBtuh,
Proposed Efficiency = 9.00 HSPF, Required Efficiency = 8.20 HSPF
Cooling Mode: Capacity = 19 kBtuh,
Proposed Efficiency = 20.20 SEER, Required Efficiency: 14.00 SEER
Fan System: None

1 SHW-1
Electric Storage Water Heater, Capacity: 20 gallons or Circulation Pump
Proposed Efficiency: 1.85 EL, %/h (P = 12 kW), Required Efficiency: 1.85 EL, %/h (P = 12 kW)

Mechanical Compliance Statement

Compliance Statement: The proposed mechanical design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed mechanical systems have been designed to meet the 2015 IECC requirements in COMcheck Version 4.1.5.5 and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Paige Faucett — 06/29/26
Name - Title — Signature — Date

COMcheck Software Version 4.1.5.5

Inspection Checklist

Energy Code: 2015 IECC

Requirements: 0.0% were addressed directly in the COMcheck software

Text in the "Comments/Assumptions" column is provided by the user in the COMcheck Requirements screen. For each requirement, the user certifies that a code requirement will be met and how that is documented, or that an exception is being claimed. Where compliance is itemized in a separate table, a reference to that table is provided.

Section # & Req ID	Plan Review	Complies?	Comments/Assumptions
C401.2 (PR27	Plans, specifications, and/or calculations provide all information with which compliance can be determined and that lists the systems and equipment and document where exceptions to the standard or other options in calculations per acceptable engineering standards and practices are claimed.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C103.2 (PR32	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the service water-heating systems and equipment and document where exceptions to the standard are claimed. Hot water system sized per manufacturer's GPM.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C406 (PR37	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the additional energy efficiency package options	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Additional Comments/Assumptions:

Section # & Req ID	Footing / Foundation Inspection	Complies?	Comments/Assumptions
C403.2.4.1 (PR57	Snow/ice melting system sensors for pavement temperature and moisture and protection systems have automatic controls installed.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Additional Comments/Assumptions:

Section # & Req ID	Plumbing Rough-In Inspection	Complies?	Comments/Assumptions
C404.5, C404.5.1, C403.5.2 (PL42	Heated water supply piping conforms to pipe length and volume requirements. Refer to section details	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.5, C404.5.1, C403.5.2 (PL42	Heated water supply piping conforms to pipe length and volume requirements. Refer to section details	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.5, C404.5.1, C403.5.2 (PL42	Heated water supply piping conforms to pipe length and volume requirements. Refer to section details	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.5, C404.5.1, C403.5.2 (PL42	Heated water supply piping conforms to pipe length and volume requirements. Refer to section details	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.6.1, C404.6.2 (PL10	Automatic time switches installed to automatically switch off the recirculating hot-water system or heat trace	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.6.3 (PL17	Pumps that circulate water between a heater and storage tank have controls that limit operation from startup to <= 5 minutes after end of heating cycle	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.6.3 (PL17	Pumps that circulate water between a heater and storage tank have controls that limit operation from startup to <= 5 minutes after end of heating cycle	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.6.3 (PL17	Pumps that circulate water between a heater and storage tank have controls that limit operation from startup to <= 5 minutes after end of heating cycle	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.6.3 (PL17	Pumps that circulate water between a heater and storage tank have controls that limit operation from startup to <= 5 minutes after end of heating cycle	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.7 (PL07	Water distribution system that pumps water from a heated-water supply system or storage tank back through a cold-water supply pipe to a fixture or appliance shall have controls Pumps within this system have controls that start the pump upon receiving a signal from the action of a user of a fixture or appliance and limits the temperature of the water entering the cold-water piping to 104°F.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Section # & Req ID	Plumbing Rough-In Inspection	Complies?	Comments/Assumptions
C404.7 (PL07	Water distribution system that pumps water from a heated-water supply system or storage tank back through a cold-water supply pipe to a fixture or appliance shall have controls that start the pump upon receiving a signal from the action of a user of a fixture or appliance and limits the temperature of the water entering the cold-water piping to 104°F.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.7 (PL07	Water distribution system that pumps water from a heated-water supply system or storage tank back through a cold-water supply pipe to a fixture or appliance shall have controls that start the pump upon receiving a signal from the action of a user of a fixture or appliance and limits the temperature of the water entering the cold-water piping to 104°F.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.7 (PL07	Water distribution system that pumps water from a heated-water supply system or storage tank back through a cold-water supply pipe to a fixture or appliance shall have controls that start the pump upon receiving a signal from the action of a user of a fixture or appliance and limits the temperature of the water entering the cold-water piping to 104°F.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Additional Comments/Assumptions:

Section # & Req ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C403.2.8 (ME147	Thermally reflective panel surfaces of sensible heating panels have emissivities >= 0.9.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.12.1 (ME157	HVAC fan systems at design conditions do not exceed allowable fan system motor nameplate hp or fan system bhp.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	See the mechanical Systems list for values.
C403.2.12.1 (ME157	HVAC fan systems at design conditions do not exceed allowable fan system motor nameplate hp or fan system bhp.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	See the mechanical Systems list for values.
C403.2.12.1 (ME157	HVAC fan systems at design conditions do not exceed allowable fan system motor nameplate hp or fan system bhp.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	See the mechanical Systems list for values.
C403.2.12.1 (ME1177	Fans have efficiency grade (FEG) >= 67, the total efficiency of the fan at the design point of operation >= 15% of maximum total efficiency of the fan	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.12.1 (ME1177	Fans have efficiency grade (FEG) >= 67, the total efficiency of the fan at the design point of operation >= 15% of maximum total efficiency of the fan	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.12.1 (ME1177	Fans have efficiency grade (FEG) >= 67, the total efficiency of the fan at the design point of operation >= 15% of maximum total efficiency of the fan	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.13 (ME717	Unenclosed spaces that are heated use only radiant heat.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.2 (ME55F	HVAC equipment efficiency verified	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	See the mechanical Systems list for values.
C403.2.4.1 (ME55F	Demand control ventilation provided for spaces > 50 sq.ft. and > 25 people per 1000 sq.ft. (Density was served by systems with air side economizers, automatic modulating outdoor air damper control, or design outdoor >3,000 cfm.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.6 (ME1157	Enclosed parking garage ventilation has automatic contaminant detection and operates at 50% of design capacity or less in 50% or less of design capacity	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.7 (ME177	Exhaust air energy recovery on systems meeting Table C403.2.7(1) and C403.2.7(2).	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Section # & Req ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C403.2.8 (ME147	Kitchen exhaust systems comply with replacement air and conditioned supply air is provided to kitchen hood ventilation requirements and maximum exhaust rate criteria.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.9 (ME685	Ducts and plenums insulated where ducts and plenums are installed in or under a slab, verification may be required by inspector during foundation inspection.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.9 (ME157	Ducts and plenums sealed based on static pressure and location.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.9.1.3 (ME112	Ductwork operating >3 in. water column requires air leakage testing.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.9.1.3 (ME112	Ductwork operating >3 in. water column requires air leakage testing.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.9.1.3 (ME112	Ductwork operating >3 in. water column requires air leakage testing.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.4.2.2.1 (ME113	Closed-circuit cooling tower within heat pump loop have either automatic bypass valve or lower leakage positive closure dampers. Open-circuit tower within heat pump loop have automatic valve to bypass all heat pump water flow around the tower. Open- or closed-circuit cooling towers used in conjunction with a separate heat exchanger have heat loss by shutting down the circulation pump on the cooling tower loop.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.4.4.4 (ME113	Multiple zone VAV systems with DDC of individual zone boxes have static pressure setpoint reset controls.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	See the Mechanical Systems list for values.
C403.4.4.4 (ME113	Multiple zone VAV systems with DDC of individual zone boxes have static pressure setpoint reset controls.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	See the Mechanical Systems list for values.
C403.4.4.4 (ME113	Multiple zone VAV systems with DDC of individual zone boxes have static pressure setpoint reset controls.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	See the Mechanical Systems list for values.
C408.2.2 (ME112	Air outlets and zone terminal devices have means for air balancing.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Section # & Req ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C403.5, C403.5.1, C403.5.2 (ME1122	Refrigerated display cases, walk-in coolers or walk-in freezers served by remote compressors and remote condensers not located in a condensing unit, have fan-powered condenser(s) that comply with C403.5.1 and refrigeration compressor systems that comply with C403.5.2.	☐Complies ☐Does Not ☐Not Applicable	

Additional Comments/Assumptions:

Section # & Req ID	Final Inspection	Complies?	Comments/Assumptions
C403.3, C408.2.5.3 (FI49	Furnished O&M manuals for HVAC systems within 90 days of system acceptance.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.2 (FI27	HVAC systems and equipment capacity does not exceed calculated loads.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.1 (FI47	Heating and cooling to each zone is controlled by a thermostat control device with humidity control device per installed humidification/dehumidification system.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.1 (FI47	Heating and cooling to each zone is controlled by a thermostat control device with humidity control device per installed humidification/dehumidification system.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.1 (FI47	Heating and cooling to each zone is controlled by a thermostat control device with humidity control device per installed humidification/dehumidification system.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.1.1 (FI42	Heat pump controls prevent supplemental electric resistance heat from coming on when not needed.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.1.2 (FI38	Thermostatic controls have a 5 °F deadband.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.2 (FI39	Temperature controls have setpoint overlap restrictions.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.3 (FI39	Each zone equipped with setback controls using automatic time clock or programmable control system.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C403.2.4.3.1, C403.2.4.3.2 (FI40	Automatic Controls: Setback to 55°F (heat) and 85°F (cool). 7-day clock, 2-hour occupant override, 10-hour backup	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.3 (FI11	Heat traps installed on supply and discharge piping of non-circulating systems	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C404.4 (FI25	All piping insulated in accordance with section details and Table C403.2.10	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Section # & Req ID	Final Inspection	Complies?	Comments/Assumptions
C404.6.3 (FI127	Controls are installed that limit the time that fossil fuel is supplied to maintain temperature of a storage tank. System return pipe is a dedicated return pipe or a cold water supply pipe.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C408.2.1 (FI28F	Commissioning plan developed by registered design professional or approved agency.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C408.2.2 (FI31F	HVAC equipment has been tested to ensure proper operation.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C408.2.2 (FI29F	HVAC control systems have been tested to ensure proper operation, calibration and adjustment of controls	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C408.2.4 (FI29F	Preliminary commissioning report completed and certified by registered design professional or approved agency.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C408.2.5 (FI17	Furnished HVAC as-built drawings submitted within 90 days of system acceptance.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C408.2.5.3 (FI43F	Air and/or hydronic system balancing report is provided for HVAC systems.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	
C408.2.5.3 (FI05F	Final commissioning report due to the building owner within 12 mo. of receipt of certificate of occupancy.	☐Complies ☐Does Not ☐Not Observable ☐Not Applicable	

Additional Comments/Assumptions:

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:
FIFTH THIRD BANK

Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON 1,900 SF

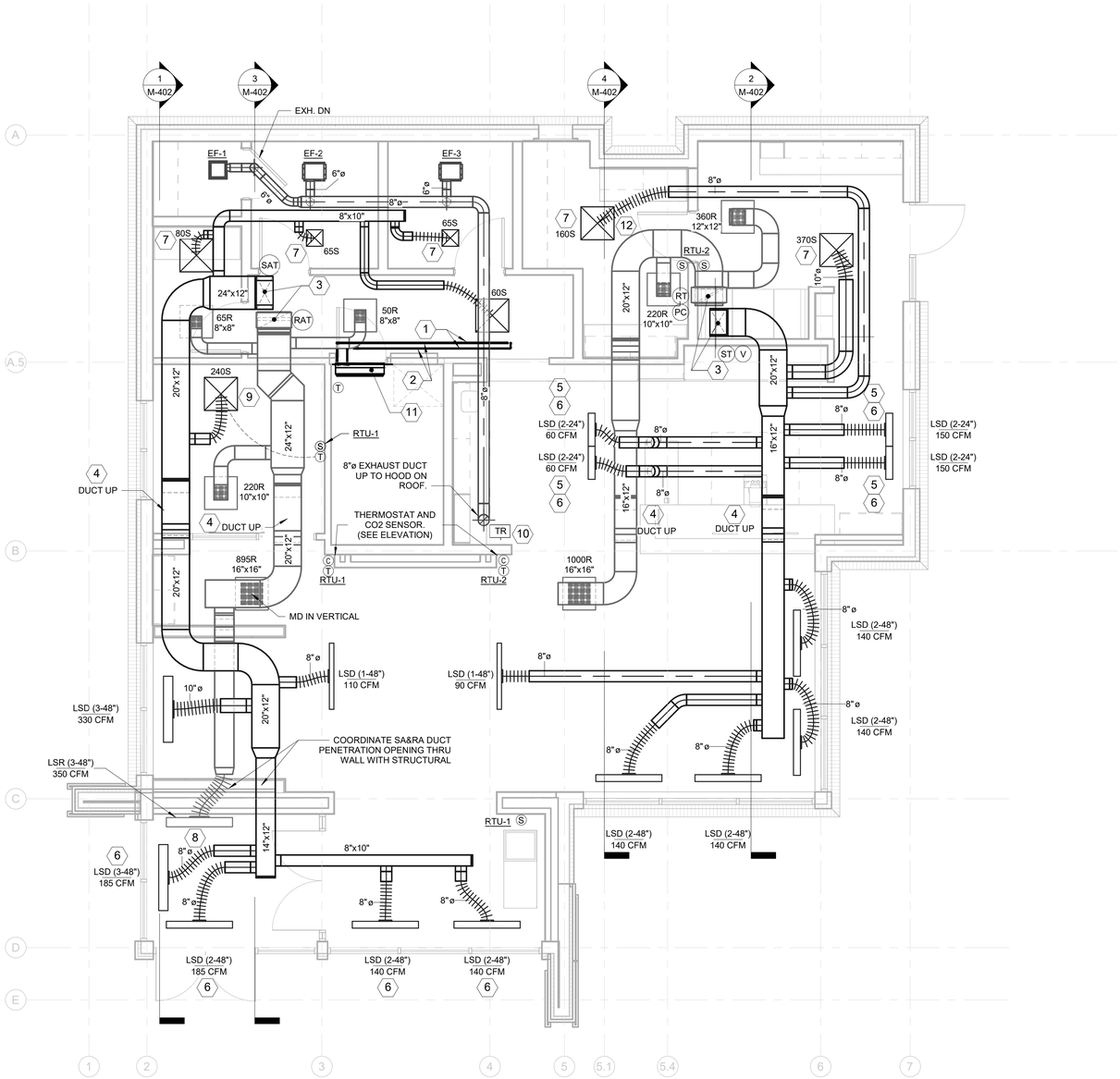
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WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
MECHANICAL VENTILATION SCHEDULES

Sheet Number:
M-102

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1 M-301 FLOOR PLAN - MECHANICAL
1/4" = 1'-0"

MECHANICAL KEY NOTES
(APPLIES TO THIS SHEET ONLY)

1. REFRIGERANT PIPING UP THRU ROOF TO OUTDOOR UNIT.
2. ROUTE REFRIGERANT PIPING AS HIGH AS POSSIBLE TO OUTDOOR UNIT ON ROOF. ROUTE AND SIZE PER MANUFACTURER'S RECOMMENDATIONS (TYP.).
3. DUCT UP AND TRANSITION TO MATCH FULL UNIT OPENING. PROVIDE FLEXIBLE CONNECTION AT UNIT. INSTALL LONG-RADIUS SWEEP ELBOW AT DUCT DROP WHERE SPACE PERMITS; OTHERWISE, PROVIDE SHORT-RADIUS ELBOW WITH TURNING VANES.
4. DUCT TO RISE ABOVE STRUCTURAL BEAM. COORDINATE WITH STRUCTURAL.
5. IN UPPER CEILING. PAINT BLACK.
6. PROVIDE REMOTE CABLE OPERATED DAMPER IN PLENUM SLOTS WITH OPERATION AT SLOT FACE.
7. PROVIDE REMOTE CABLE OPERATED DAMPER IN CEILING FOR DAMPER OPERATION.
8. LSR(3) WITH FULL SIZE SHEETMETAL PLENUM, 12" HIGH. INSIDE TO BE PAINTED FLAT BLACK. CONNECT TO DUCT TO PLENUM BOX.
9. SUPPLY DIFFUSER TO BE VAV TYPE WITH PETROLEUM BASED WAX CONTROLLED DAMPER. DAMPER SHALL CLOSE TO KEEP SPACE FROM BEING OVERHEATED ONCE ROOM TEMPERATURE IS SATISFIED (72°F ADJUSTABLE). DIFFUSER EQUAL TO PRICE ACUTHERM ST THERMA-FUSER. VAV DIFFUSER SCOPE INTENDED ONLY FOR NORTHERN PROJECTS, PREPARE TO PROVIDE BID CREDIT WHERE SCOPE IS REMOVED IN SOUTHERN PROJECTS.
10. TEMPERATURE, CO2, AND RELATIVE HUMIDITY DISPLAY EQUAL TO AUTOMATION COMPONENTS, INC. COORDINATE WITH IOT. SEE CONTROL SEQUENCE.
11. INSTALL WALL MOUNTED FAN COIL UNIT AT AN ELEVATION HEIGHT OF 8'-0" FROM FINISHED FLOOR TO BOTTOM OF UNIT.
12. INSTALL TITUS SB STEEL SECTORIZING BAFFLE IN NECK OF SUPPLY DIFFUSER SUCH THAT IT BLOCKS THE AIRFLOW FROM DIRECTLY BLOWING ONTO THE NIGHT DROP BOX.

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Client:
FIFTH THIRD BANK

Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON 1,900 SF

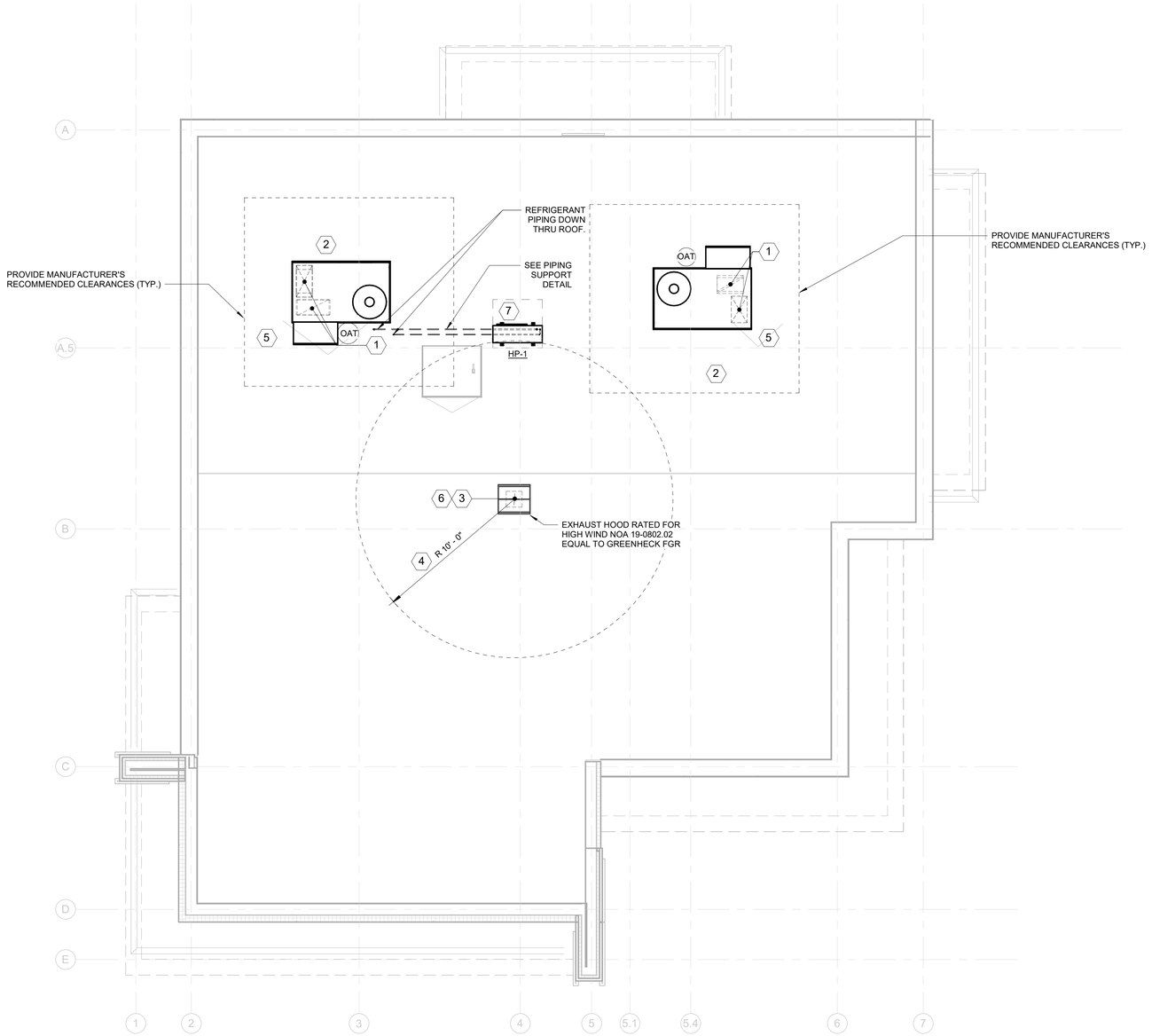
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WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
MECHANICAL PROPOSED CEILING PLAN

Sheet Number:
M-301

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1 M-302 ROOF PLAN - MECHANICAL
1/4" = 1'-0"

MECHANICAL GENERAL NOTES
(APPLIES TO THIS SHEET ONLY)

1. RTU'S TO BE SCREENED BY NEW BUILDING PARAPET WALL SO THAT ROOF-MOUNTED EQUIPMENT IS OBSCURED FROM VIEW AT GRADE. SEE ARCHITECTURAL DRAWINGS.
2. ALL EXTERIOR-MOUNTED EQUIPMENT AND SUPPORTS SHALL COMPLY WITH APPLICABLE STATE AND LOCAL PRODUCT APPROVAL REQUIREMENTS, INCLUDING FLORIDA PRODUCT APPROVAL OR MIAMI-DADE NOA WHERE REQUIRED.

MECHANICAL KEY NOTES
(APPLIES TO THIS SHEET ONLY)

1. SUPPLY AND RETURN DUCT UP TO RTU ON ROOF. COORDINATE CURB FOOTPRINT WITH STRUCTURAL BEAMS.
2. PROVIDE 1-1/2" RTU CONDENSATE DRAIN WITH P-TRAP AND REMOVABLE VENTED CAP, ROUTE TO PRIMARY ROOF DRAIN.
3. EXHAUST DUCT UP TO EXHAUST FAN ON ROOF. COORDINATE CURB FOOTPRINT WITH STRUCTURAL BEAMS.
4. MAINTAIN 10' CLEARANCE FROM ALL EXHAUST OUTLETS TO FRESH AIR INTAKES.
5. INSTALL ROOFTOP AC UNIT ON 18" WIND-RATED ROOF CURBS. CURBS SHALL BE APPROVED FOR REQUIRED WIND LOADS PER APPLICABLE BUILDING CODE. AIRSOUTH COMMERCIAL UNITS STEEL CURBS (NOA 25-0410.04) OR EQUIVALENT.
6. GRAVITY VENTILATOR SHALL BE APPROVED FOR REQUIRED WIND LOADS PER APPLICABLE BUILDING CODE. EQUAL TO GREENHECK FGR (NOA 23-1122.03) OR LOREN COOK PR-8 FOR STANDARD WIND.
7. INSTALL HP ON WIND-RATED ROOF SUPPORT. SUPPORT SHALL BE APPROVED FOR REQUIRED WIND LOADS PER APPLICABLE BUILDING CODE. EQUIVALENT TO MIAMI-TECH INC. ASMU MECHANICAL UNIT STAND (NOA 24-0117.01).

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	07/02/2026	PERMIT SET

Client:
FIFTH THIRD BANK

Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON 1,900 SF

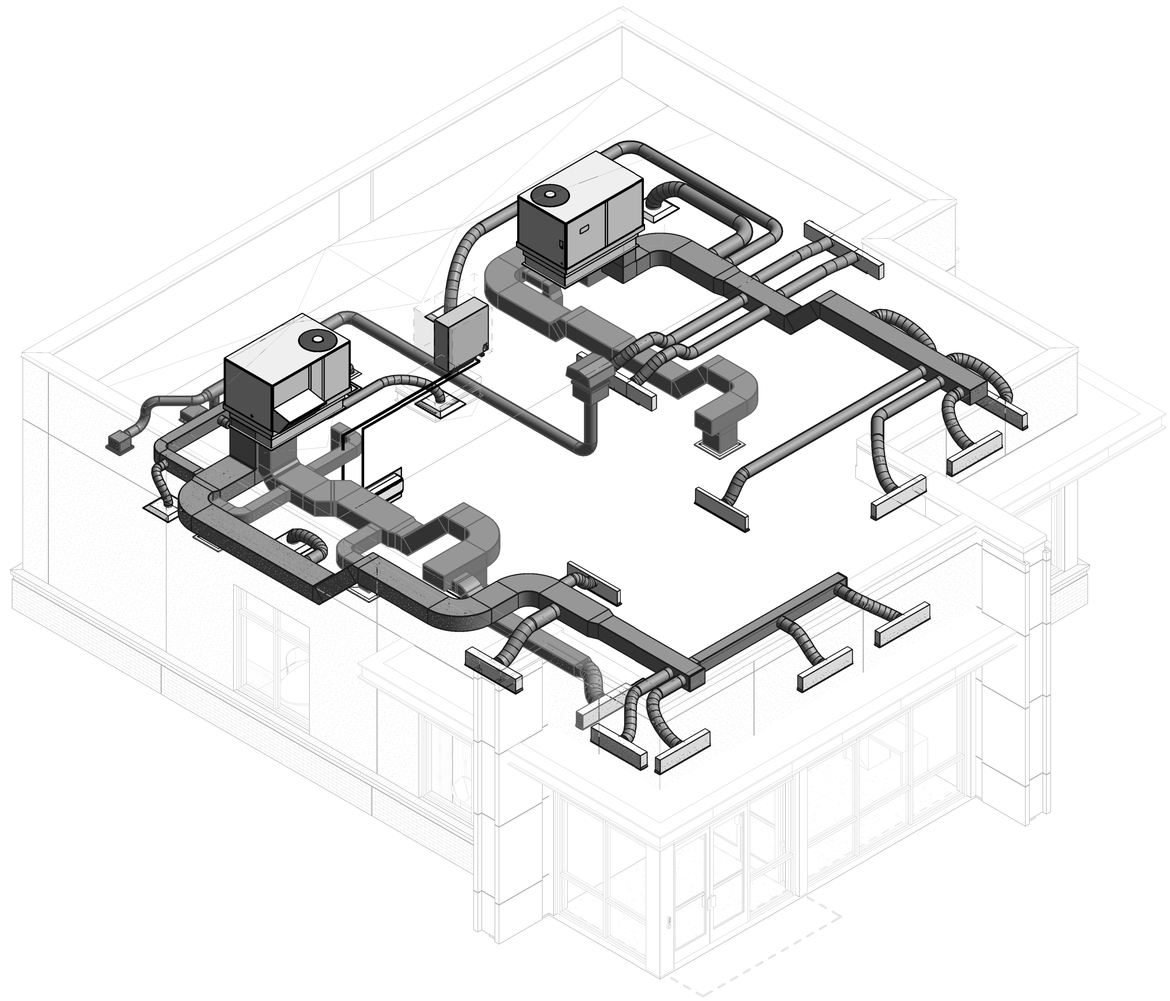
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WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
MECHANICAL PROPOSED ROOF PLAN

Sheet Number:
M-302

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1 M-401 MECHANICAL ISO PLAN

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Client:

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Project Number: 26032
Project Name:

FIFTH THIRD BANK
WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

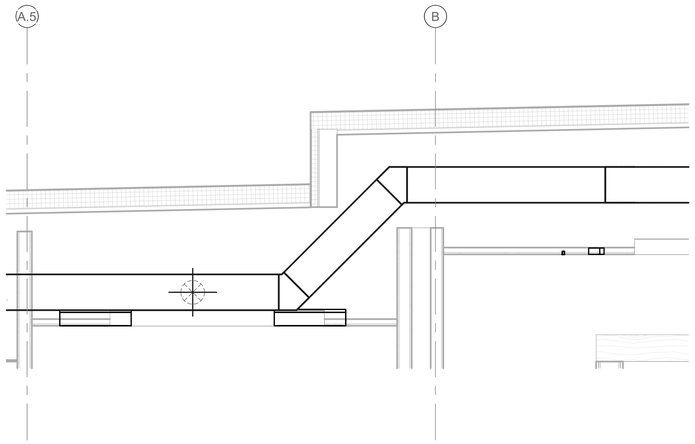
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MECHANIAL ISOMETRIC PLAN

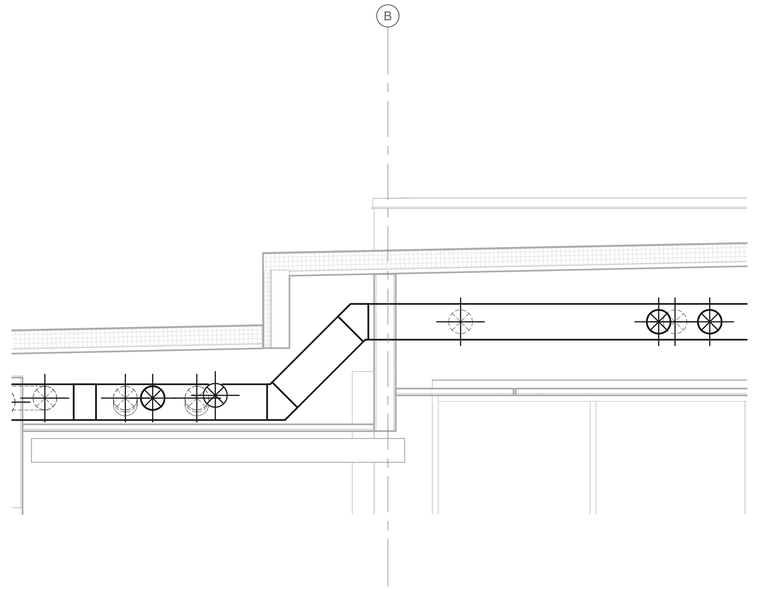
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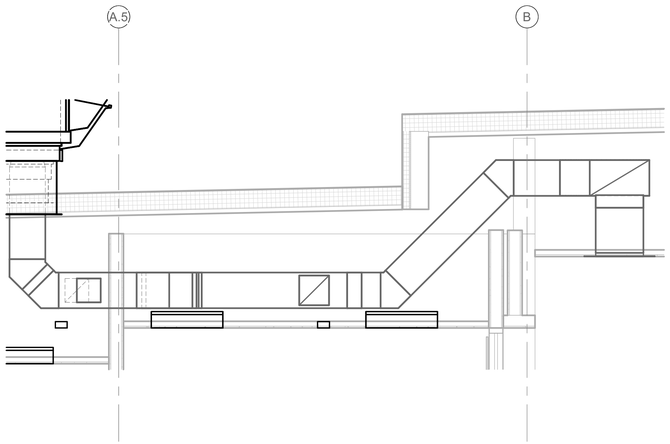
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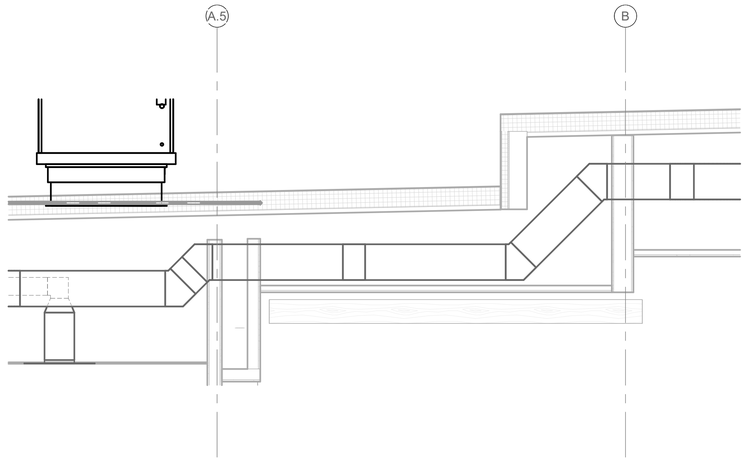
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1/2" = 1'-0"



2 M-402 RTU-2 Supply Duct Up
1/2" = 1'-0"



3 M-402 RTU-1 Return Duct Up
1/2" = 1'-0"



4 M-402 RTU-2 Return Duct Up
1/2" = 1'-0"

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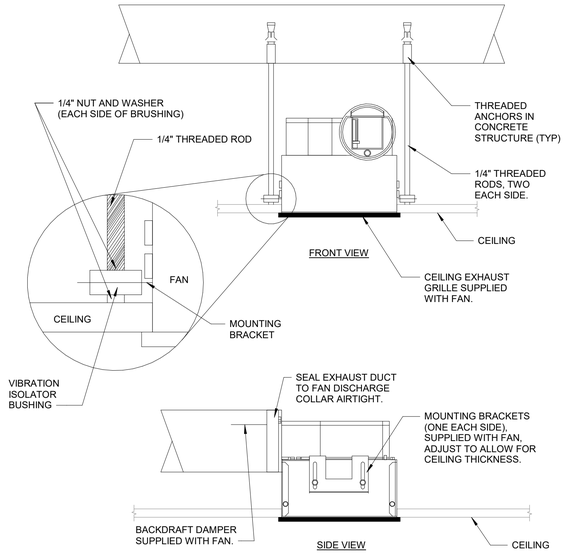
Project Number: 26032
Project Name:
FIFTH THIRD BANK
WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

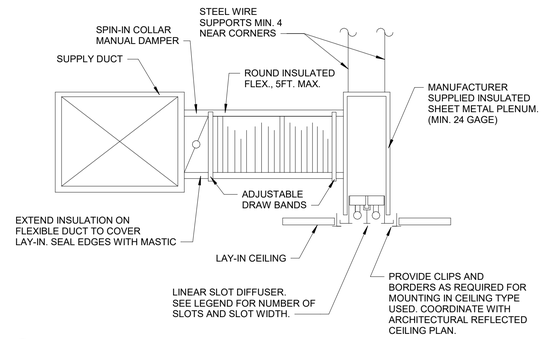
Key Plan:

Sheet Title:
MECHANICAL DUCT SECTION VIEWS

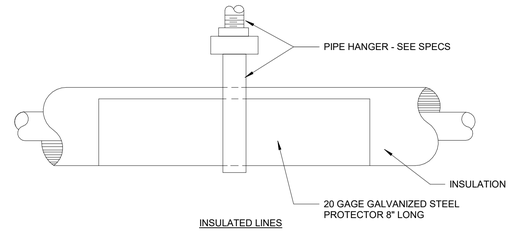
Sheet Number:
M-402

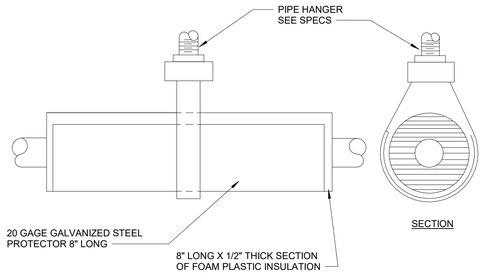


1 M-501 CEILING EXHAUST FAN DETAIL
NOT TO SCALE

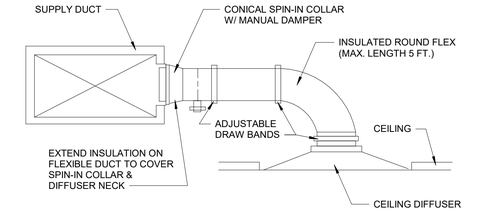


4 M-501 LINEAR SLOT CEILING DIFFUSER DETAIL
NOT TO SCALE





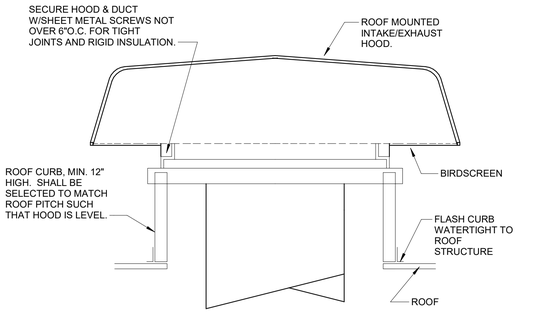
7 M-501 REFRIGERANT PIPING HANGER DETAIL
NOT TO SCALE



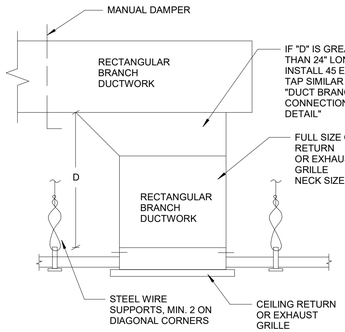
NOTES:

WHEREVER THE SUPPLY DUCT HEIGHT IS INSUFFICIENT TO CONNECT THE SPIN-IN, THE SPIN-IN MAY BE CONNECTED TO THE TOP OR BOTTOM OF THE DUCT. IF THE BRANCH DUCT MUST BE CONNECTED TO THE SIDE OF THE MAIN DUCT, USE A RECTANGULAR BRANCH DUCT CONNECTION OF EQUAL AIR VELOCITY AND TRANSITION TO ROUND DUCT. REFER TO SPECIFICATION FOR MAXIMUM TURNS IN FLEX DUCT.

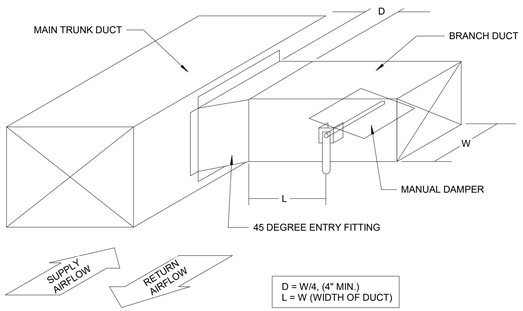
10 M-501 CEILING DIFFUSER INSTALLATION DETAIL
NOT TO SCALE



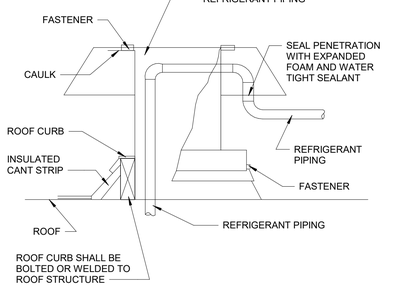
2 M-501 EXHAUST HOOD DETAIL
NOT TO SCALE



5 M-501 CEILING RETURN CONNECTION DETAIL
12" = 1'-0"

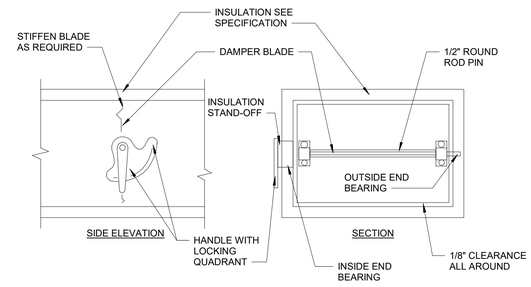


11 M-501 DUCT BRANCH CONNECTION
12" = 1'-0"



8 M-501 PIPE PENETRATION THRU ROOF
12" = 1'-0"

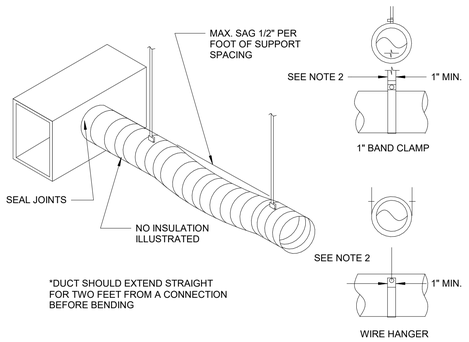
MINIMUM CONDENSATE PIPE SIZE	
AC TONS	MIN. DRAIN SIZE
0 TO 20	1"
21 TO 40	1-1/4"
41 TO 60	1-1/2"
61 TO 100	2"
101 TO 250	3"
251 & LARGER	4"



NOTES:

1. DELETE INSULATION STAND-OFF ON DUCTWORK WITHOUT EXTERIOR INSULATION.
2. DETIL SHOWS SINGLE BLAD DAMPER. DAMPER INSTALLATION SHALL BE SIMILAR FOR MULTI-BLADE DAMPERS & ROUND DAMPERS.
3. MANUAL DAMPERS SHALL BE EQUAL TO RUSKIN MD35 (FOR RECTANGULAR DUCTS) AND SHALL BE EQUAL TO RUSKIN MDRS25 (FOR ROUND DUCTS).

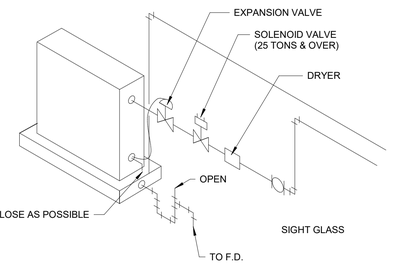
3 M-501 MANUAL DAMPER DETAIL
12" = 1'-0"



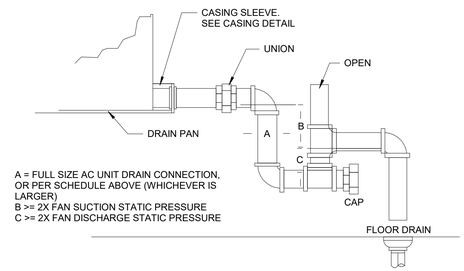
NOTES:

1. SUPPORT SYSTEM MUST NOT DAMAGE DUCT OR CASUE OUT OF ROUND SHAPE.
2. DUCTS ARE FLEXIBLE WITH EXTERNAL INSULATION AND VAPOR BARRIER JACKETING.
3. MIN. CENTER LINE BEND LINE RADIUS IS ONE DIA. (OR INSIDE RADIUS OF D/2).
4. FLEXIBLE DUCT LENGTH SHALL NOT EXCEED 5 LINEAR FEET.

6 M-501 FLEXIBLE DUCT SUPPORT DETAIL
12" = 1'-0"



9 M-501 DX COIL PIPING DETAIL
12" = 1'-0"



12 M-501 AC UNIT DRAIN TRAP DETAIL
12" = 1'-0"

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GEORGIA REGISTERED 032221 PROFESSIONAL ENGINEER GARRY M. JENKINS

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:
FIFTH THIRD BANK

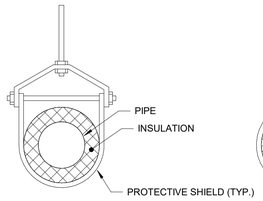
Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON 1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

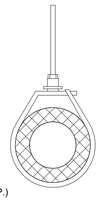
Key Plan:

Sheet Title:
MECHANICAL DETAILS

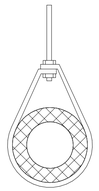
Sheet Number:
M-501



ADJUSTABLE STEEL CLEVIS ALL SIZES



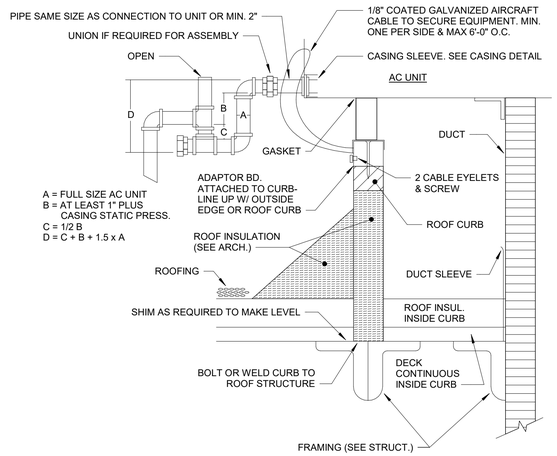
ADJUSTABLE SWIVEL PIPE RING, SPLIT RING 2" AND SMALL



ADJUSTABLE STEEL BAND HANGER, 2" AND SMALLER

PIPE SIZE	1/2"	3/4"	1"	1-1/4"	1-1/2"	2"	2-1/2"	3"	4"	6"	8"	10"
ROD DIAMTER	3/8"	3/8"	3/8"	3/8"	3/8"	3/8"	1/2"	1/2"	5/8"	3/4"	7/8"	7/8"
SPACING	6'	8'	8'	8'	10'	10'	10'	10'	10'	10'	10'	10'

1 M-502 PIPE HANGER DETAIL
NOT TO SCALE

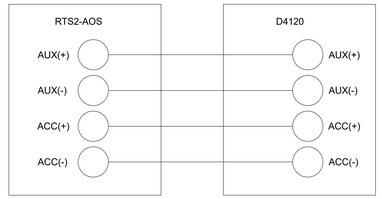


2 M-502 ROOFTOP AC UNIT DETAIL
NOT TO SCALE



1. DIMENSIONS: 4.8"WX5.3"HX1.6"
2. WEIGHT: 0.35 LB.
3. OPERATING VOLTAGE: 20-29 VDC
4. POWER REQUIREMENTS:
   A. STANDBY: 3.0 MA MAX.
   B. TROUBLE: 16.0 MA MAX.
   C. ALARM WITHOUT STROBE:30 MA MAX.
   D. ALARM WITH STROBE: 55MA MAX.
5. STRIP TERMINAL: 14AWG TO 22 AWG WIRE
6. WIRING DISTANCE: 1000 FEET FROM DUCT SMOKE DETECTOR
7. SOUNDER: 85 dBA AT TEN FEET
8. TEMPERATURE: 14 DEGREE F TO 140 DEGREE F
9. HUMIDITY: 95% RELATIVE HUMIDITY; NON-CONDENSING.
10. LISTING: UL268 AND ULC S529-02
11. STANDBY(GREEN BLINK), TROUBLE(AMBER), MAINTENANCE(AMBER BLINK) AND ALARM(RED) LEDS TO INDICATE DETECTOR STATUS
12. KEYSWITCH TO SELECT DESIRED SENSOR, ENABLE TEST/RESET BUTTON

PROVIDE WIRING AS RECOMMENDED BY EQUIPMENT MANUFACTURER



3 M-502 CONVENTIONAL MULTI-SIGNALLING ACCESSORY DETAIL
12" = 1'-0"

CONDENSER UNIT HOLD DOWN (175 MPH WIND)

A. ALL NEW CONDENSING UNITS SHALL BE PROVIDED WITH HIGH WIND PROTECTION, FOR A MINIMUM WIND VELOCITY OF 175 MPH, AS ACCEPTED FOR SOUTH FLORIDA (MIAMI DADE, BROWARD, AND PALM BEACH COUNTIES).

B. CONDENSER ANCHORING CALCULATIONS ARE BASED ON ASCE 7-16 AND THE FBC CHAPTER 16.

C. CONDENSERS SHALL BE MOUNTED ON APPROVED EQUIPMENT SUPPORT RAILS, EQUAL TO PATE MODEL ES-1. RAILS SHALL BE A MINIMUM OF 3.5" TALL, AND NOT HIGHER THAN 6" TALL. CONTRACTOR HAS THE OPTION TO USE THE PATE MODEL ES-2.

D. RAILS SHALL BE A MINIMUM OF 18 GAUGE GALVANIZED STEEL, UNITIZED CONSTRUCTION WITH INTEGRAL BASE PLATE, CONTINUOUS WELDED CORNER SEAMS, PRESSURE TREATED WOOD NAILER, COUNTER FLASHING WITH SCREWS.

E. THE RAILS SHALL BE A MINIMUM OF TWO ROOF TRUSSES.

F. THE RAILS SHALL BE ANCHORED TO THE ROOF SUPPORT TRUSSES WITH MINIMUM 3/8" ANCHOR BOLT. 3/8"∅ LAG SCREWS MAY BE USED TO ANCHOR DIRECTLY INTO THE TRUSS AT CENTERLINE OF TRUSS.

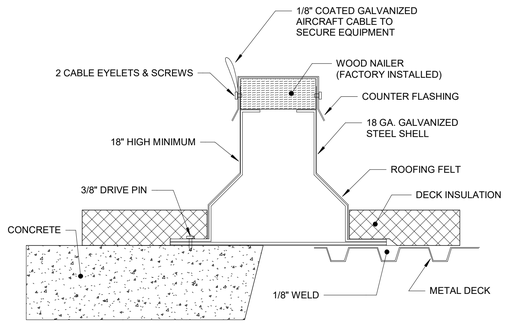
G. A MINIMUM OF TWO RAILS ARE REQUIRED TO SUPPORT A CONDENSER. MULTIPLE CONDENSERS CAN BE PLACED ON A SINGLE PAIRED RAIL SYSTEM WHERE THE RAILS SPAN MORE THAN TWO ROOF TRUSS.

H. WHERE REQUIRED, CROSS SUPPORTS THAT MAY BE REQUIRED TO SPAN THE TOP OF THE RAILS TO SUPPORT THE CONDENSING UNITS SHALL BE EQUAL TO UNIOSTRUCT P1001.

I. CONTRACTOR TO FOLLOW THE INSTALLATION GUIDELINES OF THE CONDENSER MANUFACTURER AND THE RAIL MANUFACTURER.

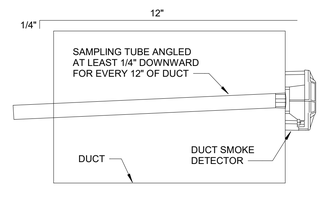
J. ALL ROOF PENETRATIONS SHALL BE SEALED WATER TIGHT.

K. RAILS TO BE FLASHED WATERTIGHT BY ROOFING CONTRACTOR.

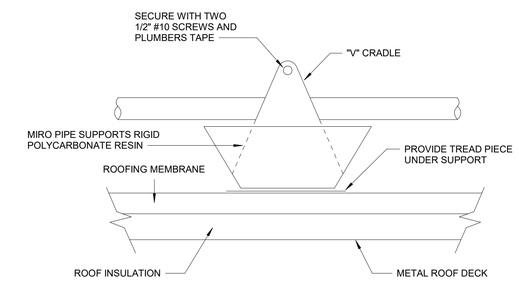


4 M-502 EQUIPMENT ROOF SUPPORT DETAIL
NOT TO SCALE

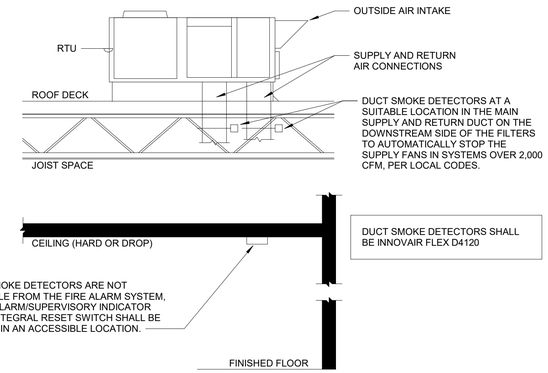
SAMPLING TUBES RECOMMENDED FOR DIFFERENT DUCT WIDTHS	
OUTSIDE DUCT WIDTH	SAMPLING TUBE RECOMMENDED
UP TO 1 FT.	DST1
1 TO 2 FT.	DST1.5
2 TO 4 FT.	DST3
4 TO 8 FT.	DST5
8 TO 12 FT.	DST10(2 PIECE)



5 M-502 TYPICAL HVAC SMOKE DETECTOR DETAIL
NTS



6 M-502 PIPING ROOF SUPPORT DETAIL
NTS



7 M-502 HVAC DUCT SMOKE DETECTOR DETAIL
NTS

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	07/02/2026	PERMIT SET

Client:

FIFTH THIRD BANK

Project Number: 26032
Project Name:

FIFTH THIRD BANK WATSON 1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:

MECHANICAL DETAILS AND CONTROLS

Sheet Number:

M-502

Category	Item	Responsible
Coordination	GC to coordinate with EC, CBRE & IOT onboarding call to review start-up process and site readiness	GC
Conduit	Wall Boxes - conduit stub-up with pull string/line (for all lighting devices and all HVAC devices) see E and M pages for details	EC
Conduit	Conduit stup up through roof curb for control wire(s) with pull string/line.	EC
Mounting	LCP Panel(s) and override(s).	EC
Power	LCP Panel(s) and override(s).	EC
Mounting	Shades Panel(s) and Override(s)	EC
Power	Shades Panel(s) and Override(s)	EC
Power	BMS control panel(s).	EC
Coordination	GC to coordinate with MC, CBRE & IOT onboarding call to review start-up process and site readiness	GC
Wiring	Thermostat wiring to HVAC system(s). Cable sized to include 2 spare conductors and leave 1 pull wire for BMS network.	MC
Mounting	POL-224 economizer controller to be fully installed and configured with accessories per IOM. Include enthalpy OA & RA sensors	MC
Mounting	Temporary thermostat control for AHU's & RTU's at the space	MC
Wiring	CO2 sensor cabling for RTU DCV control.	MC
Wiring	CO2 sensor cabling for SSAHU Controller.	MC
Start-up & Documentation	Manufacturer specified start-up and report of HVAC systems.	MC
Wiring	Install and configure 24V interface for minisplit based on manufacturer	MC
Start-up & Documentation	Submit copy or picture of start-up report to CI.	MC
Wiring	BMS and BACnet network control system wiring.	CI
Fire, Life Safety	Duct Smoke detector - See note 1.	MC
Fire, Life Safety	Fire alarm shutdown relay/connection to FACP - See note 2.	FAC
Mounting	Relief air damper and actuator.	MC
Mounting	Relief air fan or return air fan.	MC
Mounting	Outside air damper and actuator.	MC
Start-up & Documentation	As-built drawings to ME	MC
Mounting	Replace temporary thermostats with BMS thermostats (return temp to MC).	CI
Start-up & Documentation	BMS network start-up & LCP system commissioning	EC/CI
Start-up & Documentation	BMS controls & HVAC system commissioning	MC/CI
Warranty	HVAC Equipment on-board controls/sensors labor warranty.	MC
Warranty	Controls integration and user interface warranty & tech support.	CI
Warranty	HVAC equipment onboard controls/sensors parts warranty & tech support.	MANUFACTURER

MATRIX NOTES:

1. WHERE A DUCT SMOKE DETECTOR IS SHOWN, THE FOLLOWING ARE THE RESPONSIBILITIES:
   - **A.** WHERE THE BUILDING HAS A FIRE ALARM SYSTEM, THE DUCT SMOKE DETECTOR SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, AND WIRED/CONNECTED BY THE FIRE ALARM CONTRACTOR.
   - **B.** WHERE THE BUILDING DOES NOT HAVE A FIRE ALARM SYSTEM, THE DUCT SMOKE DETECTOR SHALL BE PROVIDED BY THE MECHANICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, AND WIRED/CONNECTED BY THE MECHANICAL CONTRACTOR.

2. WHERE A CONTROL MONITOR RELAY IS SHOWN FOR AN HVAC UNIT SHUTDOWN, THE FOLLOWING ARE THE RESPONSIBILITIES:
   - **A.** WHERE THE BUILDING HAS A FIRE ALARM SYSTEM, THE CONTROL MONITOR RELAY SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR, INSTALLED BY THE FIRE ALARM CONTRACTOR, AND WIRED/CONNECTED BY THE FIRE ALARM CONTRACTOR.
   - **B.** WHERE THE BUILDING DOES NOT HAVE A FIRE ALARM SYSTEM, THE CONTROL MONITOR RELAY SHALL BE PROVIDED BY THE MECHANICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, INSTALLED BY THE MECHANICAL CONTRACTOR, AND WIRED/CONNECTED BY THE MECHANICAL CONTRACTOR.

IOT BMS NOTE:

IOT SHALL PROVIDE BMS. COORDINATE WITH IOT FOR SUPPORT PRICING. CONTACT JILL MASON AT (512)-649-5142.

MECHANICAL CONTRACTOR TO SCHEDULE ON BOARDING SESSIONS WITH IOT PRIOR TO INSTALL.

1 M-503 IOT MATRIX
NOT TO SCALE

LEGEND:
CI: CONTROLS INSTALLER
EC: ELECTRICAL CONTRACTOR
MC: MECHANICAL CONTRACTOR
ME: MECHANICAL ENGINEER
GC: GENERAL CONTRACTOR
FAC: FIRE ALARM CONTRACTOR
OEM: ORIGINAL EQUIP MANUFACTURER

GENERAL NOTES

1. THE "RESPONSIBLE" COULMN BEARS THE RESPECTIVE CONTRACTOR TO PROVIDE AND INSTALL UNLESS OTHERWISE NOTED.
2. WHERE THERE IS A CONFLICT BETWEEN THE SPECIFICATIONS AND THIS MATRIX, THE MATRIX SHALL TAKE PRECEDENCE.
3. ALL CONDUIT SHALL BE INSTALLED PER NEC AND SPECIFICATIONS. ALL STUB UPS SHALL BE STUBBED UP INTO AN ACCESSIBLE CEILING. THERE SHALL BE NO CONCEALED JUNCTION BOXES

OCCUPIED MODE:
THE WALL MOUNTED PROGRAMMABLE THERMOSTAT SHALL START THE SUPPLY FAN, SUBJECT TO INTERNAL AC UNIT SAFETIES AND SMOKE DETECTOR INTERLOCK (WHERE REQUIRED). THE SPACE TEMPERATURE SENSOR SHALL CYCLE ON THE COMPRESSOR(S) TO MAINTAIN COOLING SETPOINT (74° F - ADJUSTABLE). UPON A DROP IN SPACE TEMPERATURE DURING WINTER MONTHS, THE UNIT SHALL STAGE ON THE COMPRESSOR(S) TO MAINTAIN SPACE TEMPERATURE SETPOINT (70°F - ADJUSTABLE). IF THE HEAT PUMP CANNOT SATISFY SPACE TEMPERATURE SETPOINT, THE ELECTRIC HEAT SHALL STAGE ON. DURING OCCUPIED HOURS, IF THE CO2 LEVELS RISE ABOVE 1200 PPM, THE OSA DAMPER SHALL OPEN. WHEN CO2 LEVELS DROP BELOW 800 PPM, THE OSA DAMPER SHALL CLOSE. WHERE SHOWN, AVERAGING SENSORS SHALL BE INSTALLED TO AVERAGE THE SPACE TEMPERATURE FROM MULTIPLE ZONES.

OCCUPIED HOURS TO BE DETERMINED BY THE OWNER, RECOMMENDED OCCUPIED HOURS ARE MONDAY THRU FRIDAY, 7 A.M. TO 6 P.M.

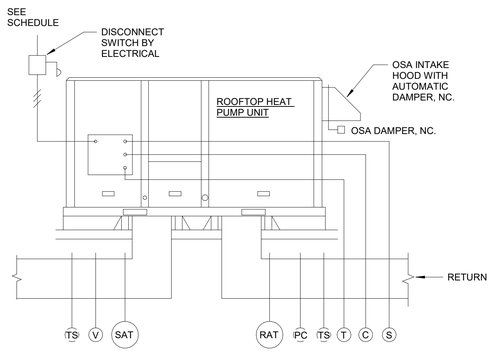
UNOCCUPIED MODE:
THE SPACE TEMPERATURE SENSOR SHALL CYCLE THE SUPPLY FAN, DX COOLING, DX HEATING, AND ELECTRIC HEAT TO MAINTAIN THE UNOCCUPIED SPACE SETPOINT (60°F HEATING / 80°F COOLING - ADJUSTABLE). THE OUTSIDE AIR DAMPER SHALL REMAIN CLOSED.

FIELD INSTALL TEMPERATURE SENSORS IN SUPPLY AND RETURN DUCTS OF ALL UNITS. SENSORS SHALL BE CAPABLE OF BEING MONITORED THRU IOT BMS/EMS.

MECHANICAL CONTRACTOR SHALL PROVIDE AND INSTALL A SINGLE TWISTED PAIR FROM THERMOSTAT TO CENTRALLY MOUNTED CONTROL PANEL.

INSTALL PARTICLE COUNTER RETURN DUCT AND VOC SENSOR IN SUPPLY DUCT (RTU-2 ONLY). COORDINATE WITH IOT.

IOT WILL SUPPLY NETWORKABLE THERMOSTAT TO BE INTEGRATED TO THE SYSTEM AND CONTROLS FOR BMS OPERATION AFTER EQUIPMENT START UP.



(T) SPACE TEMPERATURE SENSOR WITH SETPOINT ADJUSTMENT.

(C) SPACE CO2 SENSOR

(S) MONITORING SPACE TEMPERATURE SENSOR (QUANTITY AS SHOWN ON PLANS)

(SAT) FIELD INSTALLED DUCT MOUNTED SUPPLY AIR TEMPERATURE SENSOR TIED TO RTU SYSTEM AND MONITORED THRU IOT CONTROLS

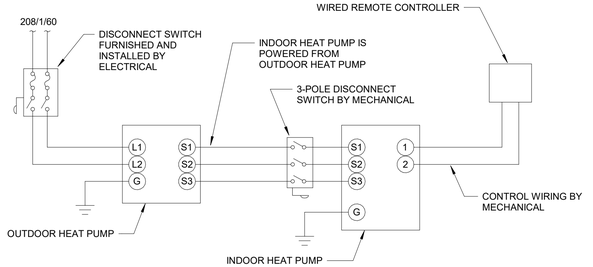
(RAT) FIELD INSTALLED DUCT MOUNTED RETURN AIR TEMPERATURE SENSOR TIED TO RTU SYSTEM AND MONITORED THRU IOT CONTROLS

2 M-503 RTU SEQUENCE AND CONTROL POINTS DETAIL
NOT TO SCALE

THE AC UNIT CONTROLLER SHALL CYCLE ON COMPRESSOR(S) TO MAINTAIN COOLING SETPOINT (74°F - ADJUSTABLE) AND HEATING SETPOINT (70°F - ADJUSTABLE).

CONTRACTOR TO PROVIDE OEM CONVENTIONAL THERMOSTAT ADAPTER WITH UNIT, PLEASE PROVIDE MITSUBISHI PAC-US445CN-1 AT PROJECT START. THE ADAPTER SHALL BE COMPATIBLE BY EITHER USING A BACNET CARD OR ALLOW INTEGRATION WITH A CONVENTIONAL THERMOSTAT.

IOT WILL SUPPLY NETWORKABLE THERMOSTAT TO BE INTEGRATED TO THE SYSTEM AND ADD CONTROLS FOR BMS OPERATION AFTER EQUIPMENT STARTUP.



3 M-503 FCU/HP SEQUENCE AND CONTROL POINTS DETAIL
NOT TO SCALE

MC TO ENGAGE A RESPRESENTATIVE FROM CARRIER (OR ALTERNATIVE MANUFACTURER) FOR START UP PRODECURES AND COORDINATION WITH IOT FOR QC PROCESS.

WORKFLOW:

1. MC TO PERFORM FULL START-UP AND CONFIGURATION OF UNIT TO FULFILL FACTORY AND JOB SCOPE REQUIREMENTS.
2. MC TO SCHEDULE AN ONBOARDING CALL WITH IOT PRIOR TO THE SITE VISIT (IOT IS OPEN FROM 8:00 AM TO 5:00 PM (CST)
   - A. PRIOR TO THE ONBOARDING CALL THE MC NEEDS TO READ THE DIGIMEP USER GUIDE
     - a. IT COVERS HOW TO USE THE APP (PG. 1-17)
       - BEST PRACTICES AND EXPECTATIONS
     - b. MEASUREMENTS TO TAKE (PG. 18)
     - c. TOOLS REQUIRED (PG. 19-20)
     - d. APP TROUBLESHOOTING (PG. 21-22)
   - B. WE WILL ALSO NEED THE FOLLOWING INFORMATION FOR THE TECHNICIAN PERFORMING THE SITE VISIT TO MAKE SURE HE/SHE HAS THE CREDENTIALS NEEDED TO LOGIN TO THE APP
     - a. FIRST AND LAST NAME
     - b. EMAIL
     - c. PHONE NUMBER
     - d. MAKE AND MODEL
3. MC TO OPEN THE DIGIMEP APP AND FOLLOW THE STEP-BY-STEP CX AND STARTUP VERIFICATION PROCESS AND SUBMIT REPORT.
4. MC AND DESIGNATED STAKEHOLDERS WILL RECEIVE A REPORT SECONDS AFTER SUBMISSION
   - A. MC WILL NEED TO REVIEW AND ADDRESS ANY DEFICIENCIES FOUND IN PERFORMANCE AND QC READINGS AND CALLOUTS.
   - B. AFTER CORRECTION OF ANY DEFICIENCIES, MC WILL OPEN THE SUBMITTED FORM, TAKE NEW READINGS TO VALIDATE UNIT IS IN PASS STATUS.
     - a. MC TO CALL IOT TO REVIEW AND VALIDATE FORM AT 512-649-5142 EXT. 7
     - b. A COMMISSIONING TECH WILL REVIEW THE FORM WITH YOU TO MAKE SURE YOU ARE GOOD TO GO.

NOTE: ALL OF THESE READINGS AND ASSET DATA, PERFORMANCE DATA AND PICTURES WILL CONFIRM QUALITY OF INSTALL AND WILL BE A PERMANENT ASSET RECORD THAT WILL VALIDATE SCOPE AND THAT THE RESULTANT PERFORMANCE IS ACCEPTED.

4 M-503 ONBOARDING START-UP REVIEW MEETING
NTS

IOT BMS NOTE:

IOT SHALL PROVIDE BMS. COORDINATE WITH IOT FOR SUPPORT PRICING. CONTACT GERRIANNE FAULHABER.

EMAIL: GERRIANNE@IOFTEAM.COM
PHONE: (224)213-6908

MECHANICAL CONTRACTOR TO SCHEDULE ON BOARDING SESSIONS WITH IOT PRIOR TO INSTALL.

STARTUP MEETING NOTE:

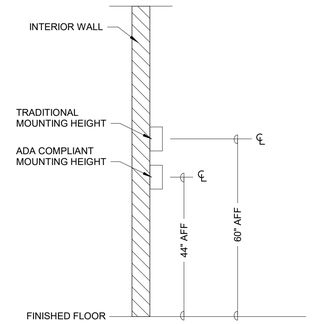
GC TO COORDINATE AN ON BOARDING EMAIL WITH THE EC, MC, CBRE PM. THE GC NEEDS TO ATTEND THE MEETING.

LOW VOLTAGE NOTE:

MC TO REVIEW IOM MANUAL FOR LOW VOLTAGE ROUTING. LOW VOLTAGE ROUTING TO BE RUN PER IOM AND NOT ON SAME CONDUIT AS HIGH VOLTAGE.

THERMOSTAT MOUNTING NOTES:

1. ALL THERMOSTATS SHALL BE INSTALLED AT 44" AFF UNLESS NOTED OTHERWISE.
2. MAINTAIN 16" FROM ADJACENT EXTERIOR WALLS. WHERE INSTALLATION IS REQUIRED ON EXTERIOR WALL, PROVIDE INSULATED MOUNTING PAD FOR BASE PLATE.
3. DISTANCE FROM CORNERS AND IF OTHER SENSORS NEARBY TO BE SIDE BY SIDE.
4. DISTANCE FROM WALL CORNERS TO BE 6" MINIMUM.
5. REFRAIN FROM INSTALLING THERMOSTAT NEAR ANY HEATING OR COOLING SOURCES; SUCH AS EQUIPMENT, WINDOWS, OR UNDER SUPPLY DIFFUSERS.
6. ENSURE THERMOSTAT IS INSTALLED IN CORRECT ORIENTATION AND ACCORDING TO ALL MANUFACTURER INSTALLATION INSTRUCTIONS.
7. COORDINATE FINAL LOCATION WITH IN FIELD CONDITIONS, MAINTAIN CLEARANCE FROM NEARBY SWITCHES AND OTHER DEVICES AS NEEDED.
8. IF OTHER SENSORS ARE LOCATED NEARBY INSTALL SIDE BY SIDE.
9. J-BOXES ARE TO BE USED ON ALL MONITORING SENSORS, THERMOSTATS AND CO2 SENSORS.



5 M-503 WALL THERMOSTAT MOUNTING DETAIL
NTS

6/30/2026 4:55:05 PM

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:
FIFTH THIRD BANK

Project Number: 26032
Project Name:
FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:
MECHANICAL CONTROLS DETAILS

Sheet Number:
M-503